DRIVE SYSTEM INSTRUCTIONS

GEK-65150B
for
DS3820DTM

Directo-Matic® PLUS

Drive Test Module

General Electric Company
Drive Systems Operations
Salem, Virginia U.S.A.

GENERAL ELECTRIC

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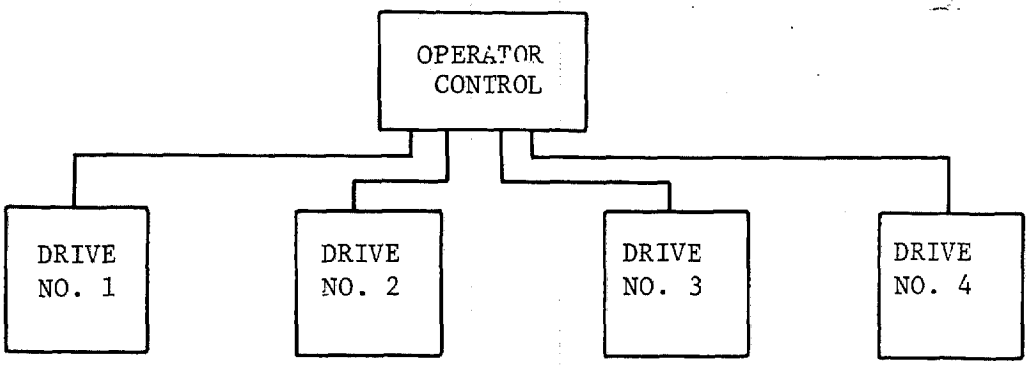
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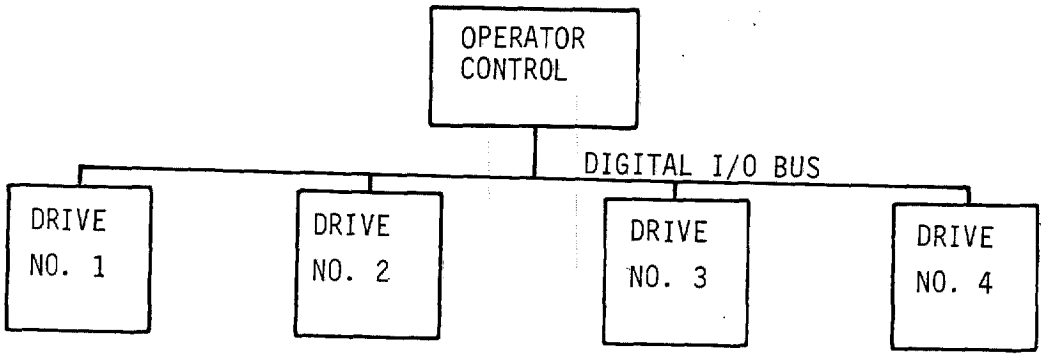
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1. PURPOSE OF EQUIPMENT

The Drive Test Module (DTM) is a maintenance aid to facilitate tune-up and troubleshooting of Specific Direct Control level DIRECTO-MATIC® Plus single and multi-drive systems. A DTMB module is required for maintenance on Digital I/O components. A typical Direct I/O system contains an operator control function which maintains direct control over one or more drives. A typical Direct I/O system is configured as follows:



A typical Digital Bus I/O system also contains an operator control function which communicates with all drives via a shared digital bus. A typical simplified bus I/O system is configured as follows:



1. PURPOSE OF EQUIPMENT (continued)

The DTM direct I/O controls feature the following fucntions:

a. Two adjustable test references
b. A digital voltmeter readout
c. A drive suicide control
d. A motor contactor control
e. A logic point input
f. A logic point readout

Additionally, the DTM contains a protective power-up circuit which insures that the drive is not started or stopped as a result of powering-up the DTM.

The major functions of the DTM digital I/O bus are to:

a. Actively read digital I/O points.
b. Actively write digital I/O points.
c. Passively monitor digital I/O data transfers.
d. Detect parity errors.

2. PHYSICAL DESCRIPTION

The DTM is enclosed in a locking, aluminum case with carrying handle. A storage space for one test cable and one AC power cord is located inside of the top cover of the carrying case. Figure 1 shows the DTM and supplied cables.

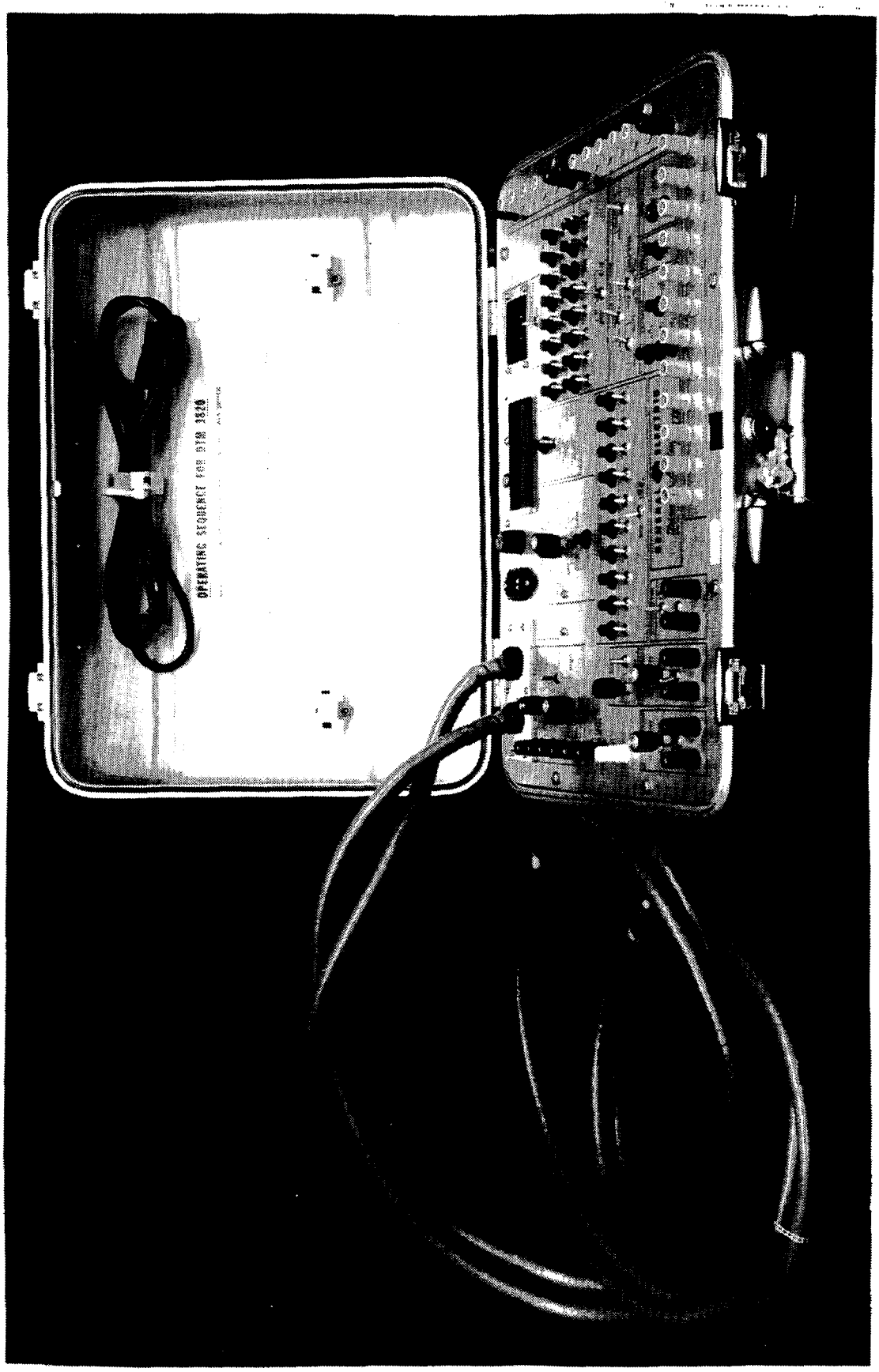
Dimensions:

Width - 14 1/4 inches (36.2 cm)

Length - 18 1/4 inches (46.4 cm)

Depth - 7 1/4 inches (18.4 cm)

Weight: 21 pounds (3 kg.)



(DSF52984)

Figure 1. Drive Test Module and Cables

3. ADDITIONAL EQUIPMENT REQUIRED

A wire (not supplied) is required to connect to the 5-way binding post on the DTM. This wire must be supplied by the user. An oscilloscope may also be required to provide voltage-time relationship information for test/troubleshooting purposes. A multimeter may be necessary for continuity tests, high voltage (>19.99 VDC) measurements and for current tests.

4. CONNECTING DTM TO EQUIPMENT UNDER TEST

Two cables must be used to connect the DTM to the equipment under test. They are the:

a. Maintenance test cable
b. AC power cable

Prior to connecting the DTM to the equipment under test, the associated drive should be disabled even though safety interlocks are used in the DTM.

If the DTM is to be connected to a drive system, the following steps must be performed:

a. Stop the drive by suiciding.

b. Disconnect any jumper leads (that may have been connected for a previous test) between the DTM and the enclosure under test.

CAUTION

IF STEPS a AND b (ABOVE) ARE NOT PERFORMED, INADVERTENT MOVEMENT OF THE DRIVE MOTOR CAN OCCUR.

To connect the DTM to the enclosure under test:

a. Connect the AC power cable.

b. Connect the maintenance test plug between the enclosure under test and the DTM (See Figure 2 for an illustration of connector mating).

To mate the maintenance test plug to a receptacle:

(1) Rotate the DTM plug actuator lever fully counterclockwise (unlocked position).

(2) Carefully engage the maintenance test plug to the test receptacle.

(3) Firmly hold the maintenance test plug against the test receptacle and rotate the actuator lever fully clockwise 1/4-turn to vertical position.

To remove the maintenance test plug:

(1) Set DTM POWER switch to OFF position.

(2) If connected to a drive, ensure the drive is suicided.

(3) Rotate DTM plug actuator 1/4-turn counterclockwise, and remove the plug.

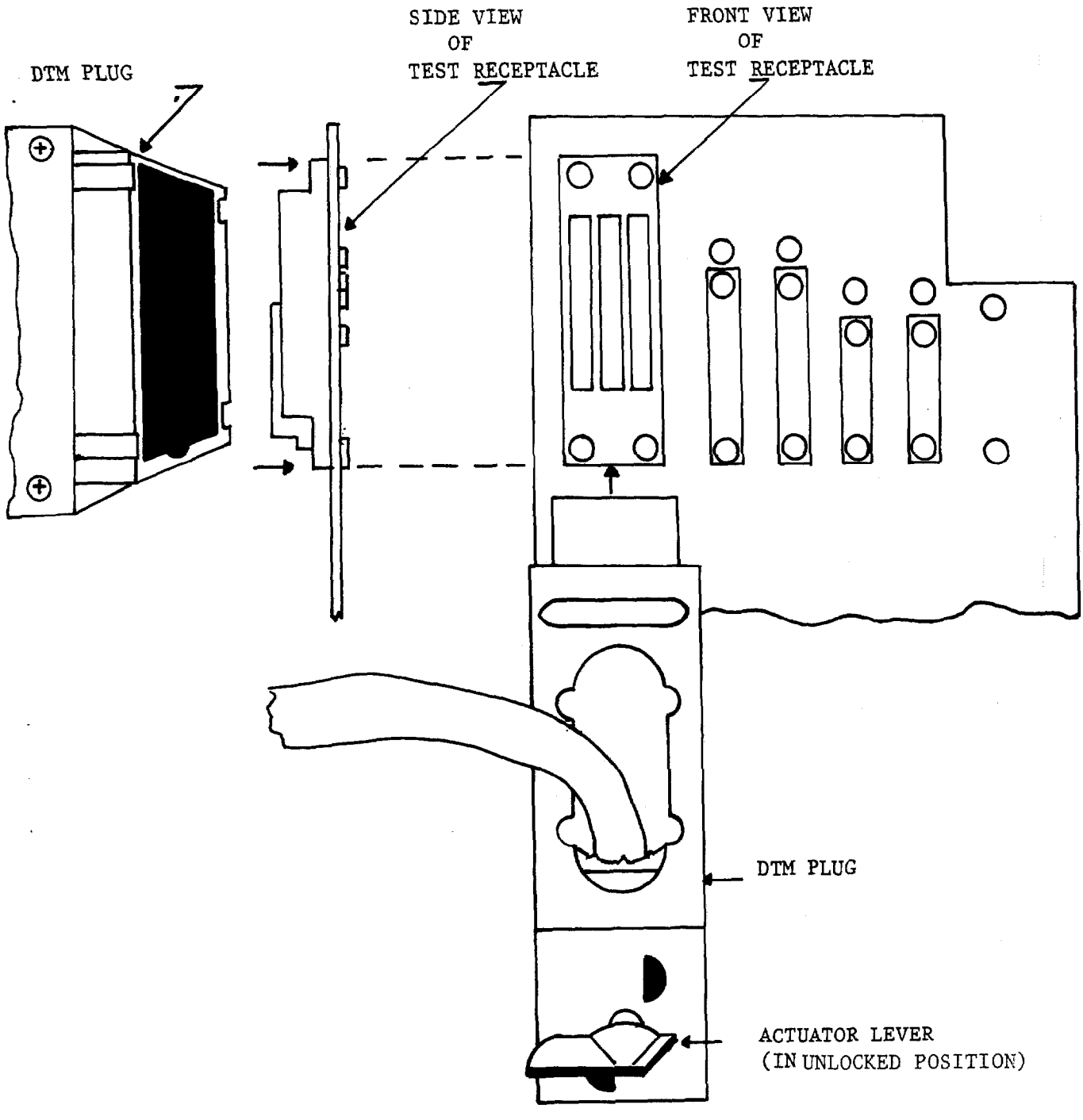


Figure 2. Engaging DTM Plug With Test Receptacle

5. OPERATING CONTROLS

The operating controls are segmented into three major parts: Encircled numbers in Figure 3 designate the paragraph numbers which describe the particular control or indicator discussed. (1) Direct I/O Controls, are mostly on the left-hand side of the test set panel, but also include the SIGNAL ACCESS test jacks and TEST POINT OUTPUT test jacks on the right hand side of the panel. The Direct I/O controls are applicable to both the DTMA and DTMB modules. (2) The Digital I/O controls are located on the right hand side of the panel (excluding the SIGNAL ACCESS and TEST POINT OUTPUT jacks. The digital I/O controls are found on DTMB modules only. (3) The POWER ON/OFF switch, located near the center of the panel, is common to both Direct I/O and Digital I/O segments of the panel and is therefore used on both DTMA and DTMB modules. See Figure 3.



A. POWER OFF/ON Switch - This switch controls the turn-on and off of the DTM. A light-emitting diode adjacent to the POWER switch illuminates when the DTM is turned on.

Since the DTM is capable of starting and stopping a drive, an interlock circuit within the DTM prevents powering up the DTM unless the switches listed below are set for a "power-up" condition. Note that the correct switch positions are identified by a box around the correct "power-up" switch position.

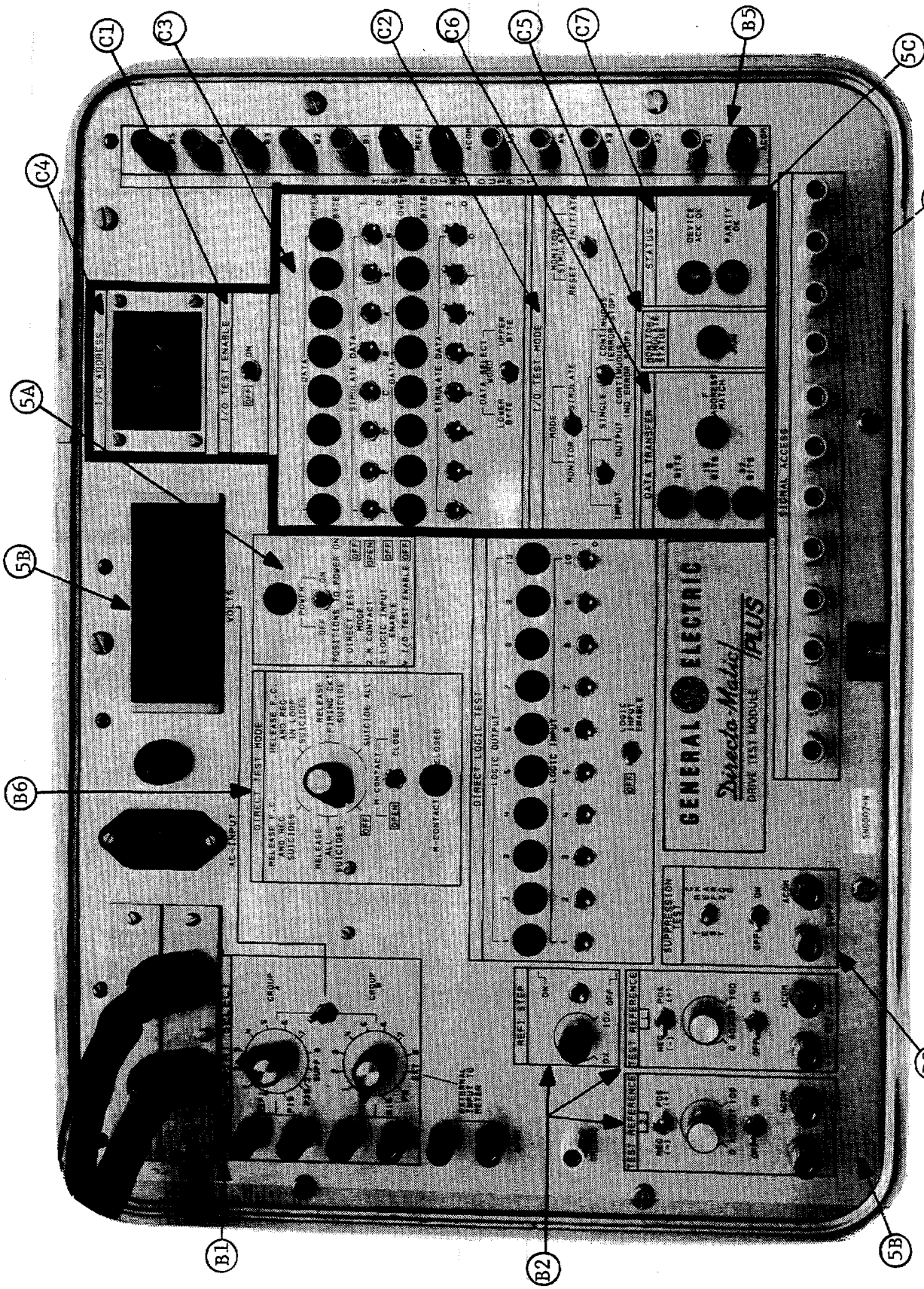
Power-Up Switch Position

Switch	Position
DIRECT TEST MODE	[OFF]
M CONTACT-OPEN	[OPEN]
LOGIC INPUT ENABLE	[OFF]
I/O TEST ENABLE	[OFF]

NOTE

No interlock exists for powering-off the DTM. Power is removed from the DTM by setting the POWER switch in the OFF position.

B. Direct I/O Controls. The direct I/O control of the DTM (the total area of the front panel outside the heavy block line as shown in Figure 3) provides the facilities required to monitor and manipulate key analog signals and to place the drive into test mode conditions. The DTM also allows the operator to read and write logic signals contained within the drive, i.e., permissives, suicides, etc. The signals to be monitored must be transmitted by hard wire.



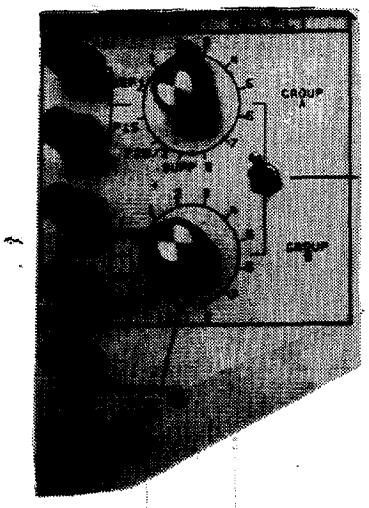
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Figure 3. Drive Test Module Front Panel

5. OPERATING CONTROLS (continued)

Jumpering of signals between the DTM and system under test is minimized because most key signals can be monitored and controlled at the DTM front panel.

Figure 4 illustrates major test functions and signal connections between the Direct I/O circuitry and drive electronics. A 3 1/2-digit, digital voltmeter is mounted at the top of the direct I/O panel. This meter is used for the readout of analog signals. Light-emitting diodes (LEDs) allow logic level output signals to be monitored. Adjustable test reference signals are available for injection into key analog or digital inputs within a drive. Furthermore, one test reference provides the ability to produce an adjustable 0-10% stepped reference output.



B1. TEST POINT SELECT and Digital Voltmeter. The TEST POINT SELECT and digital voltmeter allow the operator to monitor key analog signals originating in the drive electronics as well as power bus voltages. The input to the meter is selected by two 12-position selector switches. These switches connect the meter to one of 24 inputs divided into two groups of 12 designated GROUP A and GROUP B. A toggle switch located between the TEST POINT SELECT switches selects the switch (GROUP A or GROUP B) to be in control.

Each TEST POINT SELECTOR switch is capable of connecting any one of eight signals (labelled 1 through 8) from the drive to the digital voltmeter. The actual signals connected are dependent upon requisition design specifications. Power bus signals (P5, P15, N15 and P28) from the drive to the digital voltmeter are also selectable. Three test references (REF 1, REF 2 and SUPP) are also available for monitoring. An external input (EXT INPUT TO METER) is available to allow monitoring of any desired signal within meter limits. The maximum allowable voltage across the meter is $\pm$19.99 VDC.

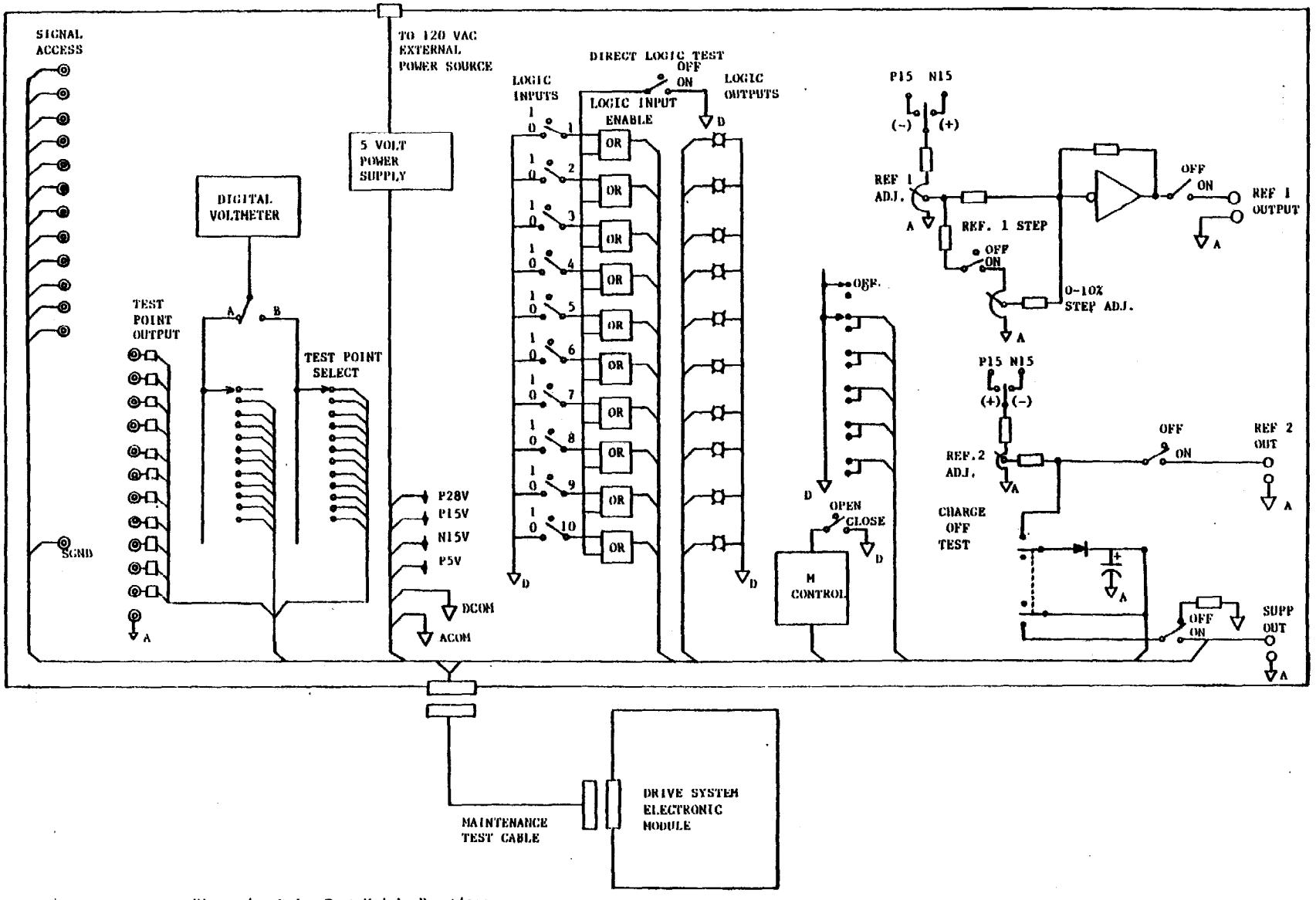
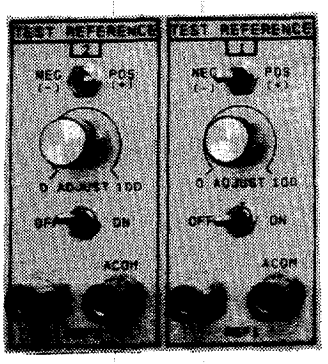


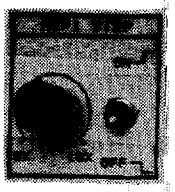
Figure 4 Drive Test Module Functions

5. OPERATING CONTROLS (continued)

There are two sets of test jacks labelled POINT SELECT A and B. The "A" jacks enable monitoring any signal selected by GROUP A METER POINT SELECTOR. Similarly, the POINT SELECT B jacks enable monitoring any signal selected by the GROUP B METER POINT SELECTOR. Similarly, the POINT SELECT B jacks enable monitoring any signal selected by the GROUP B METER POINT SELECTOR.

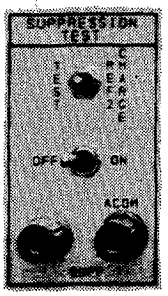


B2. General Purpose References. The purpose of the REF 1 and REF 2 outputs is to provide voltage reference sources for various test points throughout the drive electronics. These reference voltages are developed between the REF 1/REF 2 test jacks and the ACOM test jacks. These voltages are adjustable by the ADJUST potentiometers to ±13.5 VDC. The polarity is selectable by the POS-NEG switches. Use of these signals requires jumpering between the REF 1/REF 2 output jacks and drive electronics maintenance test points or SIGNAL ACCESS jacks.



Key signals are available through SIGNAL ACCESS test jacks. (See Paragraph B4). Jumpering of signals between the DTM and drive electronics is minimized by the availability of the SIGNAL ACCESS test jacks and REF 1/REF 2 outputs. The output of the REF 1 and REF 2 circuits is enabled by ON-OFF switches located adjacent to the REF 1/REF 2 jacks.

A REF 1 STEP circuit contains a step adjustment knob and an ON/OFF switch. The step adjustment control allows the amplitude of the step to be adjusted continuously between 0 and 10% of REF 1 value. The REF 1 output is stepped by the ON/OFF switch. The ON/OFF switch is debounced.



5. OPERATING CONTROLS (continued)

B3. SUPPRESSION TEST. The purpose of the SUPPRESSION TEST reference is to provide a high-current, short duration pulse to test the firing circuit suppression circuitry. There are two ways of delivering the voltage pulse from the DTM to the firing circuit suppression circuitry as follows:

a. Jumpering between the SUPP test jack output and the high voltage board burden resistor. Ensure that the positive test jack from the suppression output is connected to the positive side of the burden resistor on the high voltage board. A connection to the negative (-) side of the burden resistor is unnecessary because the DTM ACOM and burden resistor negative side are common.

b. Mechanically (and electrically) connecting the DTM to the test receptacle. (No external connections are required.)

NOTE

One of the two test connection methods described above is required and further discussed in the literature supplied with the requisition.

The ON/OFF switch enables or disables the suppression circuit test jack output. After the suppression circuit is enabled, the suppression test voltage level is adjusted using the REF 1 potentiometer. The TEST/OFF/CHARGE switch must be depressed to CHARGE position in order to allow the voltage to build to the required level. To perform a suppression test, proceed as follows:

1. Set the SUPPRESSION TEST ON/OFF switch to ON position.

2. Set the TEST POINT SELECT-GROUP A/GROUP B switch to the GROUP A position.

3. Set the TEST POINT SELECT-GROUP A selector switch to SUPP position.

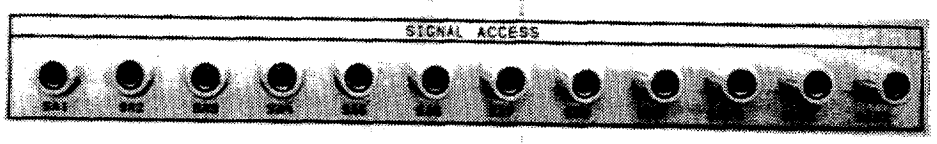
4. Depress the SUPPRESSION TEST/OFF/CHARGE switch to the CHARGE position.

5. Monitoring the digital voltmeter, adjust the REF 2 ADJUST potentiometer for the required voltage level.

5. OPERATING CONTROLS (continued)

NOTE

A capacitor in the SUPPRESSION TEST circuit charges up to the voltage level adjusted in step 5, above. If the desired voltage level is exceeded by inadvertently rotating the REF 2 ADJUST potentiometer too far clockwise, the excess charge on the capacitor cannot be reduced by rotating the REF 2 ADJUST potentiometer counterclockwise. In the event an excessive charge is developed, the capacitor charge can be removed as follows: Set the SUPPRESSION TEST OFF/ON switch to the OFF position and momentarily set the TEST/OFF/CHARGE switch to the TEST position. Thereafter repeat steps 1 thru 5 above to obtain the desired SUPP voltage amplitude setting.



B4. SIGNAL ACCESS Test Jacks. These test jacks provide a local access capability for injecting test signals into drive electronics, thereby avoiding the necessity of externally jumpering connections. There are twelve separate SIGNAL ACCESS jacks connected to various test points within the drive electronics. These test points are normally drive test inputs. (See requisition information provided to determine the specific connecting points within the drive electronics.)

5. OPERATING CONTROLS (continued)

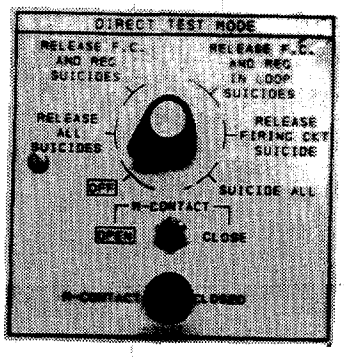
B5. TEST POINT OUTPUT Test Jacks. These test jacks allow plug-in of various types of recording equipment, i.e. brush recorders, strip line recorders, etc. These jacks allow the operator to monitor up to ten outputs simultaneously. Additionally, each of the TEST POINT OUTPUT jack signals can be individually selected by the METER POINT SELECT switches for display on the digital voltmeter. The selection of RECORDER signals for display on the digital voltmeter is determined by the following METER POINT SELECT switch settings:



TEST POINT SELECT SWITCH (See Paragraph B1)

GROUP A Selector Switch Position	TEST POINT OUTPUT Jack	GROUP B Selector Switch Position	TEST POINT OUTPUT Jack
REF 1	REF 1	1	B1
1	A1	2	B2
2	A2	3	B3
3	A3	4	B4
4	A4		
5	A5		

The TEST POINT OUTPUT jack signals, located within the drive electronics, depend upon requisition requirements and are defined in data supplied with each requisition. An exception to this is the REF 1 and ACOM test jacks which are always fixed.



B6. DIRECT TEST MODE. The DIRECT TEST MODE selector switch commands the drive system to go into one of 5 test modes available within its logic. The test modes are as follows:

1. SUICIDE ALL - Suicides firing circuit, regulator (both inner and outer regulator loops), and reference manipulation card.
2. RELEASE FIRING CIRCUIT SUICIDE - Releases firing circuit suicide condition and retains all other suicide conditions.
3. RELEASE F.C. AND REG. IN LOOPS SUICIDES - Releases firing circuit and regulator inner loop suicides.

5. OPERATING CONTROLS (continued)

4. RELEASE F.C. AND REG. SUICIDES - Releases firing circuit and regulator inner and outer loop suicides.

5. RELEASE ALL SUICIDES - Releases reference manipulation, regulator and firing circuit and suicides.

6. OFF - Opens all connections to drive analog test mode circuits.

M CONTACT Switch - This switch allows the operator to open or close the motor contactor of the drive system only when the DIRECT TEST MODE switch is not in the OFF position.

C. Digital I/O Controls. The digital I/O portion of the DTM provides the capability of monitoring and manipulating key digital I/O signals. Refer to Drive and Direct Control Equipment Instruction Book (GEK-61680) for detailed information relative to direct control bus and digital I/O characteristics. The digital I/O portion of the DTM is shown (outlined with a heavy black line) in Figure 3. The digital I/O section of the DTM provides two basic operating modes for test and troubleshooting of the digital I/O bus system. These modes are: (1) SIMULATE mode which permits the DTM to simulate Programmable Master Controller (PMC) transfers between the PMC and I/O cards, or (2) MONITOR mode where the DTM monitors data transfers between the PMC and specific direct control level I/O cards. Additionally, the SIMULATE or MONITOR functions of data transfers may be defined for input or output cards. Thus, the DTM can be put into one of four specific test modes as follows:

MODE	FUNCTION
SIMULATE-OUTPUT	DTM transfers the data selected by data switches to an output card.
SIMULATE-INPUT	DTM inputs data from an input card to its DATA LED indicators.
MONITOR-OUTPUT	DTM displays PMC to output card data transfers on its DATA LED indicators.
MONITOR-INPUT	DTM displays input card-to-PMC data on its DATA LED indicators.

The data flow associated with each of the above operating modes is illustrated in Figure 5.

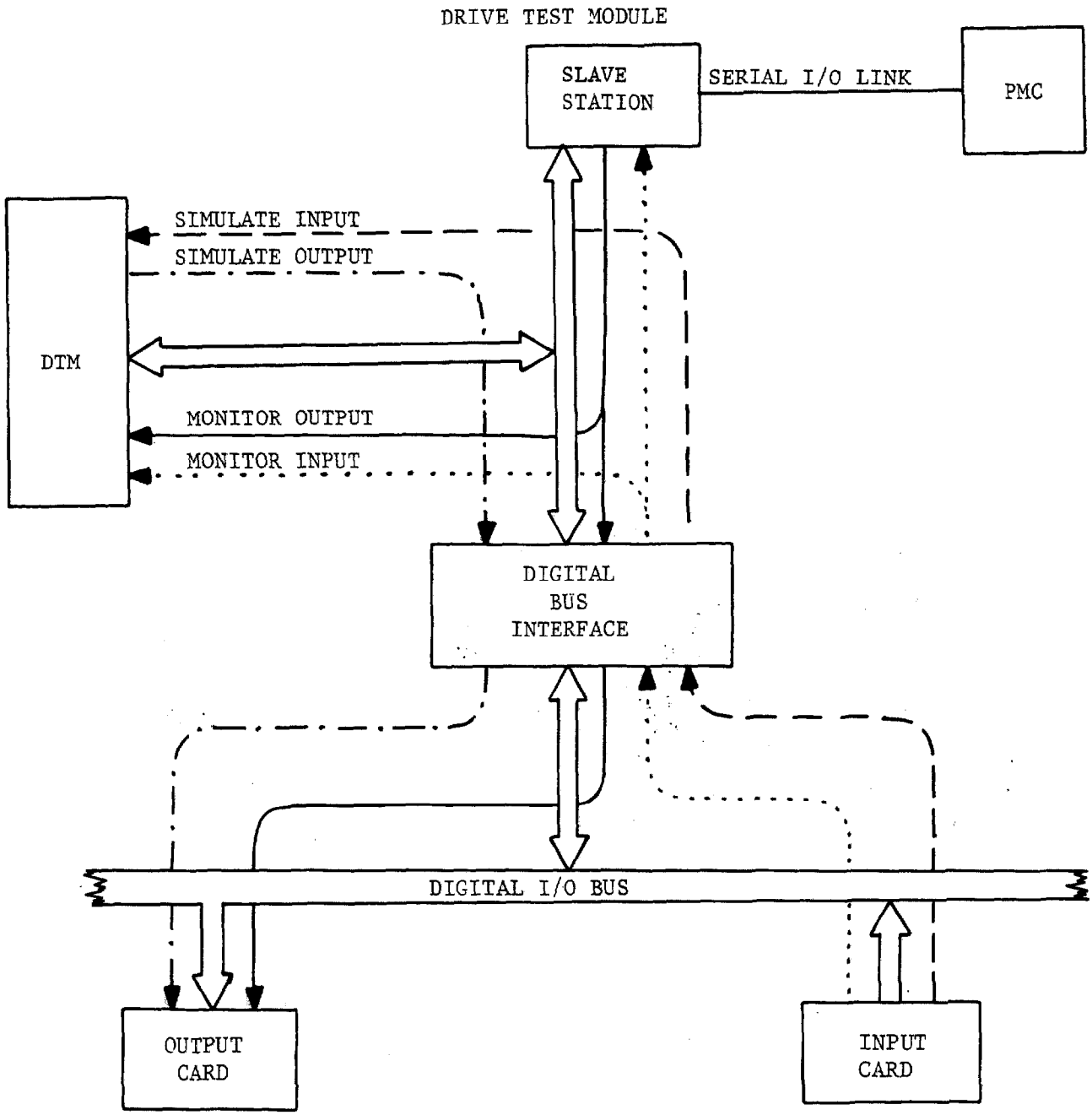


Figure 5. Data Flow Associated with Digital I/O Modes of Operation

5. OPERATING CONTROLS (continued)

The digital I/O bus address that the DTM will respond to for either the SIMULATE or MONITOR mode is selected by a thumbwheel switch. Data which is removed from the digital I/O bus by the DTM is internally latched and indicated by the 16 LED DATA indicators. Data which is transferred to the digital I/O bus by the DTM is determined by the position of sixteen DATA switches. Secondary indicators are provided to indicate any error(s) associated with the MONITOR or SIMULATE mode.

The DTM is connected to the digital I/O bus at the output of the slave station (DS3800HLSA card) via the maintenance test connector. Refer to Figure 6.

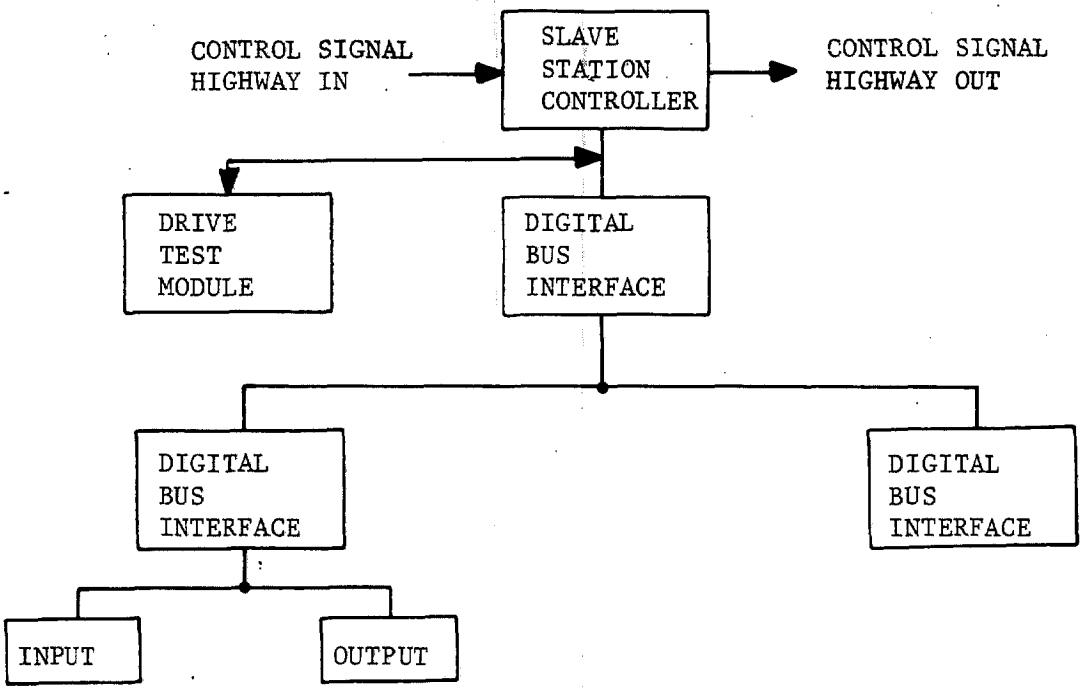
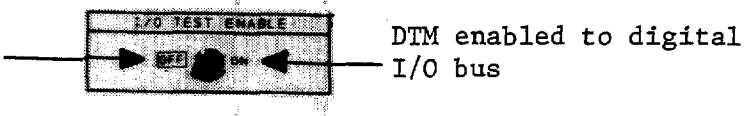
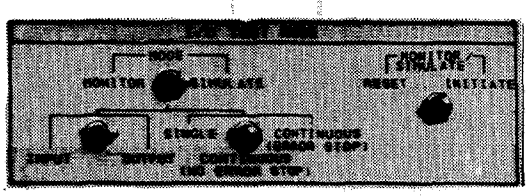


Figure 6. Digital I/O Bus Connected to DTM.

5. OPERATING CONTROLS (continued)



C1. I/O TEST ENABLE. In the ON position, the I/O TEST ENABLE switch enables the digital I/O controls of the DTM to communicate thru the digital I/O bus. In the OFF position, the DTM is isolated from the bus and all display indications are extinguished.



C2. I/O TEST MODE. The I/O test mode switches shown above establish how the DTM will function relative to the digital I/O bus. There are four basic test modes which are selected by the MONITOR/SIMULATE and INPUT/OUTPUT switches.

A summary of I/O TEST MODE switch functions is as follows (additional information is given on pages 24 and 25):

MODE-MONITOR Switch - Used to monitor PMC - TO/FROM - I/O card data transfer.

MODE-SIMULATE Switch - Used to simulate PMC-to-I/O card data transfer.

INPUT Switch Position - Selects input data transfer for either a MONITOR or SIMULATE condition.

OUTPUT Switch Position - Selects output data transfer for either a MONITOR or SIMULATE condition.

SINGLE Switch Position - Enables a test consisting of a single data transfer.

CONTINUOUS (ERROR STOP) Switch Position - Enables a test consisting of continuous data transfers. A "stop test" occurs if a data transfer error is detected.

CONTINUOUS (NO ERROR STOP) Switch Position - Enables a test consisting of continuous data transfers. A test will not stop if a data transfer error is detected.

5. OPERATING CONTROLS (continued)

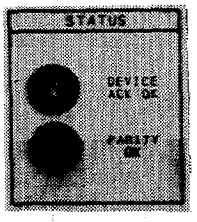
MONITOR/SIMULATE RESET Switch Position - Stops an on-going MONITOR or SIMULATE test, clears DATA TRANSFER indicators and presets DATA indicators.

MONITOR/SIMULATE INITIATE Switch Position - Used to initialize DTM for test (when depressed) and start test (when released).

SIMULATE Mode Operation - The DTM operates as an active device on the digital I/O bus while in SIMULATE mode. The DTM will simulate Programmable Master Controller (PMC) - initiated data transfers to an I/O card. To accomplish this, the DTM takes control of the digital bus.

In the SIMULATE mode, data is either transferred from the DTM to an output card or data is received by the DTM from an input card. The flow of data, either to or from the DTM is selected by the INPUT/OUTPUT switch. When the SIMULATE OUTPUT mode is selected, the DTM transfers the value selected on the SIMULATE DATA switches to the addressed output card. When the mode is SIMULATE INPUT, the data from the input lines is displayed on the DATA indicators.

Several error conditions may occur in the course of attempting to simulate a data transfer. The existence of an error is made evident to the user by the status indicator:

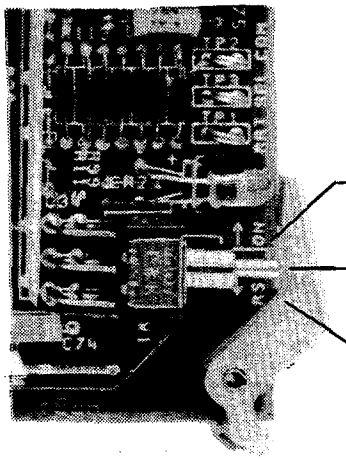


For example, if the DEVICE ACK OK indicator is off, a read or write to a non-existent card has been attempted. See paragraph C7 for causes of error indications.

If an error occurs, the DTM will abort the execution of the SIMULATE data transfer while the test mode condition is SINGLE or CONTINUOUS (ERROR STOP). However, if the mode is CONTINUOUS (NO ERROR STOP), the DTM will continuously attempt to SIMULATE data transfers until the mode is stopped by a reset from the MONITOR/SIMULATE switch or from the digital I/O bus reset.

I/O TEST MODE Operation. Digital bus interface cards (e.g. HBDA, HPBA, etc., plus the slave station card (HLSA)) have a front edge-mounted, 3-position toggle switch (See Figure 7) which places the card into one of three modes.

5. OPERATING CONTROLS (continued)



"ON" - Card is enabled to permit data to flow through the DBI.

"MAINT" - The card isolates the data link side from the I/O card side but does not generate a digital I/O bus reset.

"RST" - Same as "MAINT" but a reset is generated to digital I/O bus.

BOTTOM

(ICF51441)

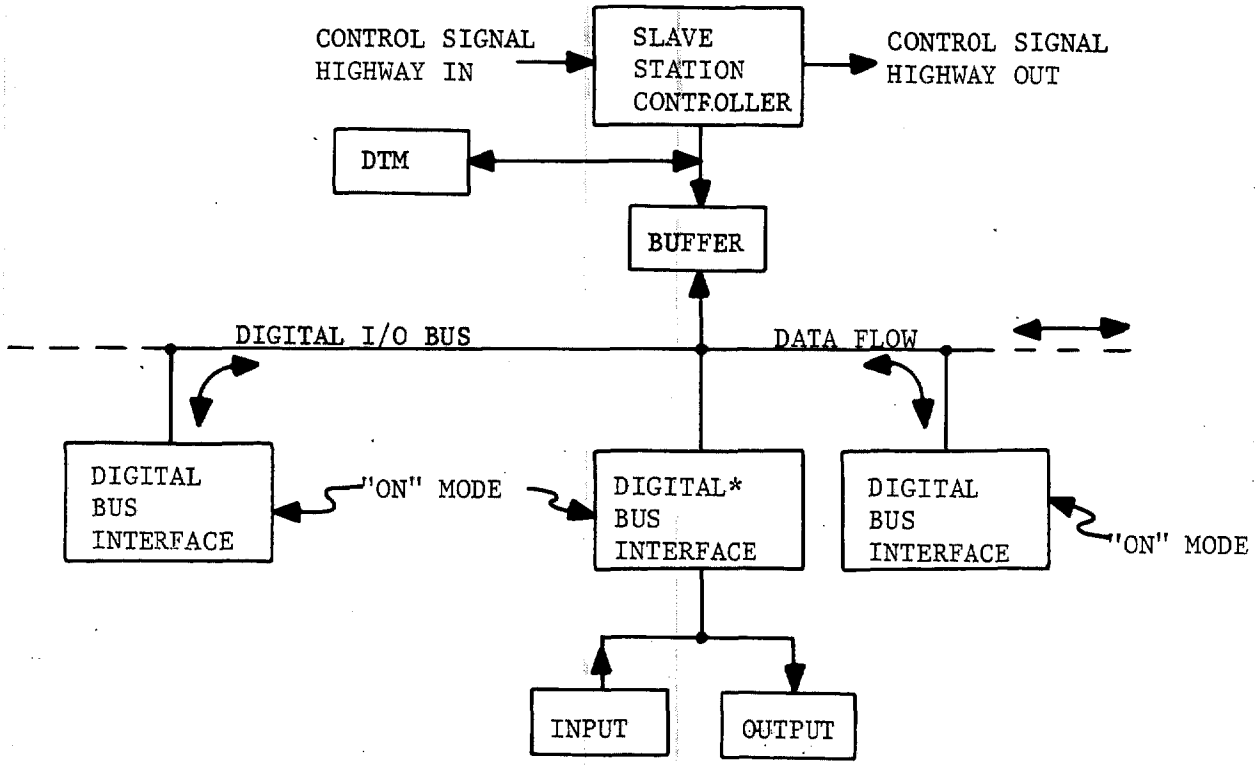
Figure 7. Part of Digital Bus Interface Card DS3800HPBA.

When operating in the SIMULATE mode, it is important to prevent the PMC from accessing the same digital I/O bus that the DTM is connected to. To do this, the slave controller card (DS3800HLSA) that is electrically closest to the DTM and is positioned between the DTM and PMC should be placed in the maintenance mode. In this mode, the slave station controller card will isolate the digital I/O bus from the control signal highway, thus preventing the PMC from accessing the I/O cards associated with the slave station (See Figure 8).

NOTE

When monitoring data transfers (using the MONITOR mode), it is necessary that the digital bus interface (DBI) mode switch be in the "ON" position and that the slave station controller be in the non-maintenance mode. As such, the DTM can monitor data transfers to the slave station and through the respective DBI card.

5. OPERATING CONTROLS (continued)



*NOTE: Programmable Master Controller is inhibited from accessing I/O cards associated with this slave station.

Figure 8. System Configuration with DTM Enabled for SIMULATE Mode.

MONITOR Mode Operation. While in the MONITOR mode, the DTM is operating as a passive device. The DTM monitors the undecoded address lines. If a match exists between the I/O ADDRESS thumbwheel value and the digital I/O bus address lines along with the appropriate control signals, the data present on the digital I/O bus data lines is latched into the DTM.

The MONITOR mode operation is performed in parallel to the PMC-I/O card data transfers being monitored. The monitoring of the digital I/O bus is receive-only and therefore does not interfere with PMC-I/O card data transfers. Moreover, in this mode, the PMC cannot detect the presence of the DTM on the digital I/O bus.

5. OPERATING CONTROLS (continued)

The data latched into the DTM is displayed on the DATA indicators. The MONITOR mode can be set to monitor 8 bits (either upper or lower byte) or 16 bits (a word). If, for some reason, the PMC initiates a data transfer to an unassigned I/O address (an address where no I/O card is designated) the DTM will not latch any data.

INPUT Switch Position. When the INPUT/OUTPUT switch is in the INPUT position, the I/O test mode is associated with "input" cards. Per the MONITOR/SIMULATE MODE switch, the DTM will either:

a. MONITOR PMC data transfers from input cards, or

b. SIMULATE data transfers to an input card.

OUTPUT Switch Position. When the INPUT/OUTPUT switch is in the OUTPUT position, the I/O test mode is associated with the output cards. Per the MONITOR/SIMULATE MODE switch, the DTM will either:

a. MONITOR PMC data transfers to output cards, or

b. SIMULATE data transfers to an output card via switches.

SINGLE Switch Position. This switch position enables the DTM to either monitor or simulate one data transfer. After the single data transfer has been completed, the DTM stops the test.

CONTINUOUS (NO ERROR STOP) Switch Position. This switch position keeps the DTM enabled to MONITOR or SIMULATE data transfers after a test has been initiated, the operating mode may be terminated one of two ways:

a. Setting the MONITOR/SIMULATE switch in the RESET position, or

b. Generating a reset from the digital I/O bus (typically by throwing the digital bus interface card and slave station controller card (DS3800HLSD) toggle switch to the RESET position).

If a data transfer error is encountered (e.g., data transfer to a non-existent device, resulting in a device acknowledge error) the DTM will remain enabled in the SIMULATE/MONITOR mode. (See paragraph C7 for additional causes of error conditions.)

CONTINUOUS (ERROR STOP) Switch Position. This switch function is identical to the CONTINUOUS (NO ERROR STOP) switch function with one exception: a data transfer error will stop a MONITOR/SIMULATE test. Stopping a test due to an error is signaled to the operator by the MONITOR/SIMULATE STATUS RUN indicator and one of the two STATUS OK indicators being extinguished. The STATUS indicators alert the operator as to the type of error detected. (See paragraph C7.)

5. OPERATING CONTROLS (continued)

INITIATE Switch Position. When the MONITOR/SIMULATE switch is set to the INITIATE position, the DTM initializes its logic and resets error indications to execute a test. Upon returning the MONITOR/SIMULATE switch to the center (off) position, the DTM initiates a MONITOR/SIMULATE test. After a test is started, the INITIATE function of the MONITOR/SIMULATE switch is electrically disabled for the duration of the test.

NOTE

The initiation of a test mode will be inhibited if a reset condition (i.e., 1BCRRT=1 or 1BCFRT=1) exists on the digital I/O bus. To quickly determine if a bus reset is present, put the DTM into a CONTINUOUS, MONITOR mode test. If the MONITOR/SIMULATE STATUS indicator does not light, a bus reset condition exists. This is also a good indication that the DTM is in working order.

RESET Switch Position. When the MONITOR/SIMULATE switch is set to the RESET position, one of the following actions results:

1. The current SIMULATE or MONITOR test is stopped.

2. DATA TRANSFER indicators (if lit) will be turned off.

3. If the MONITOR mode is selected, eight data indicators will be turned on per the DATA SELECT switch. The DATA indicators are affected as follows:

DATA SELECT WORD Switch Position	DATA Indicators Turned On
LOWER BYTE	0-7
WORD	None
UPPER BYTE	8-F

5. OPERATING CONTROLS (continued)

Either the upper byte, lower byte or word (both bytes) may be enabled for a given test. The set of indicators to be enabled is determined by the position of the DATA SELECT switch. Either the upper byte or lower byte DATA indicators, as applicable, will illuminate to indicate the presence of logic "1" data.

SIMULATE DATA Switches. Sixteen SIMULATE DATA switches (0 thru 7 and 8 thru F) are used in the SIMULATE-OUTPUT test mode. The SIMULATE DATA switches determine the data value that will be transferred to an output card from the DTM. The DATA SELECT switch determines whether the upper byte, lower byte or both bytes (a word) are transferred during a SIMULATE OUTPUT operation. When the upper or lower byte is selected, only that byte is affected by a SIMULATE OUTPUT operation. However, when using the DTM to access a four-bit output card, it is not possible to manipulate the four bits of one card without also affecting the four bits that make up the byte.

The up position of the SIMULATE DATA switches selects a logic "1" and the down position a logic "0".

NOTE

A logic "1" causes an output card to turn on the respective output device, e.g. a light, relay, contact, etc. A logic "0" causes an output card to turn off respective output devices.

DATA SELECT Switch. This switch determines which SIMULATE DATA switches and DATA indicators are applicable for a SIMULATE or MONITOR mode data transfer. The three positions of this switch function as follows:

LOWER BYTE - The lower 8 data indicators or SIMULATE DATA switches (numbers 0 thru 7) will be enabled.

UPPER BYTE - The upper 8 data indicators or SIMULATE DATA switches (numbers 8 thru F) will be enabled.

WORD - All DATA indicators or SIMULATE DATA switches will be enabled.

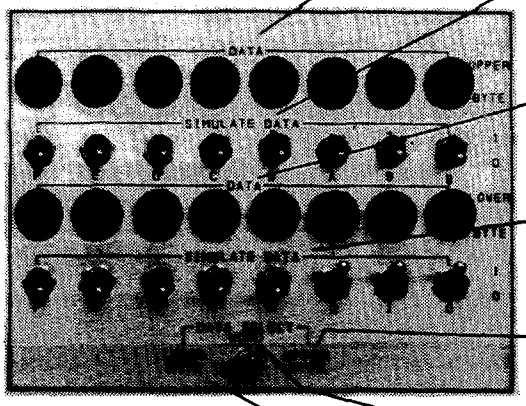
5. OPERATING CONTROLS (continued)

C3. SIMULATE DATA Switches, DATA Lights and DATA SELECT WORD switch. The switches and indicators within this group of controls provide the user with the following capabilities (See sketch, below):

1. The DATA indicators display monitored or simulated data.

2. The SIMULATE-DATA switches allow the user to select the data value to be transferred to an output card on a SIMULATE-OUTPUT mode of operation. The switches are disabled in all other test modes.

3. The DATA SELECT switch allows the user to define which set(s) of DATA indicators and SIMULATE DATA switches to be enabled for a SIMULATE or MONITOR test.



DATA indicators provide visual display of upper byte (bits 8-F) data transfers.

SIMULATE DATA switches allow selection of upper byte data for a SIMULATE output.

DATA indicators provide visual display of lower byte (bits 0-7) data transfers.

SIMULATE DATA switches allow selection of lower byte data for a SIMULATE output.

DATA SELECT-UPPER BYTE - enables UPPER BYTE switches and/or indicators.

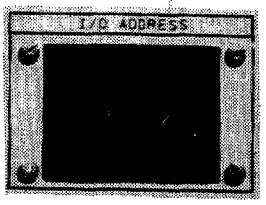
DATA SELECT-WORD-enables UPPER and LOWER BYTE switches and/or indicators.

DATA SELECT-LOWER BYTE - enables LOWER BYTE switches and/or indicators.

DATA Indicators: Two sets of 8 LED indicators, labeled 0 thru F, are available to review the results of the following test operations:

1. SIMULATE - INPUT
2. MONITOR - INPUT
3. MONITOR - OUTPUT
4. SIMULATE - OUTPUT

5. OPERATING CONTROLS (continued)



C4. I/O ADDRESS THUMBWHEELS. The I/O ADDRESS thumbwheels (See Figure 1) specify the 2-digit hexadecimal address that ia to be monitored or the address of the source/destination of a SIMULATE data transfer. When selecting an I/O address, the operator should be aware of the following:

(1) A thumbwheel switch read-out of addresses 0 thru FF is selectable. However, since the most significant digit line of the thumbwheel switch is not used, only addresses 00 thru 7F are valid.

(2) The DTM monitors or simulates undecoded address lines. This means that the I/O ADDRESS switch value plus the DATA SELECT switch is used by the DTM for the undecoded address that will be monitored or simulated. (See Figure 9.)

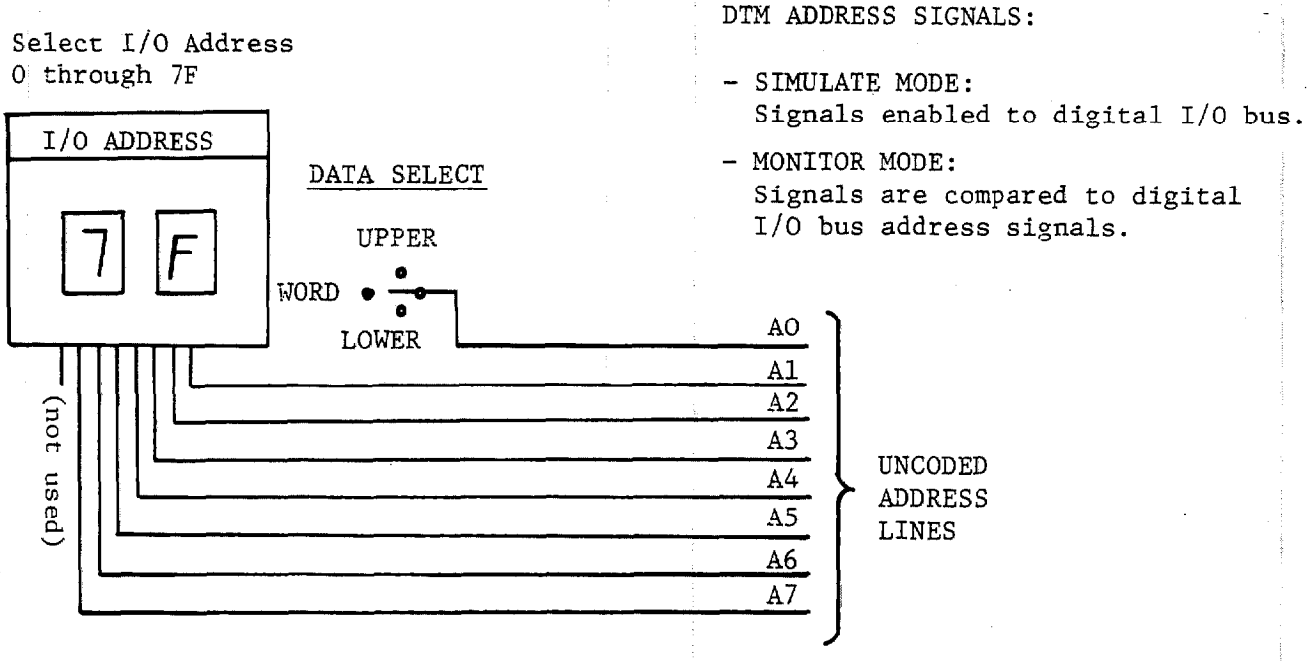
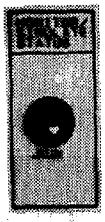


Figure 9. DTM Address Circuitry

5. OPERATING CONTROLS (continued)

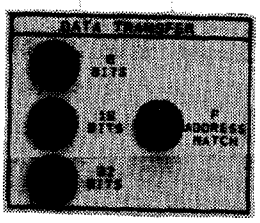


C5. MONITOR/SIMULATE STATUS. A single LED indicator (Shown above) is provided to indicate when a MONITOR or SIMULATE test is in progress. If a MONITOR or SIMULATE test is stopped (e.g. by a RESET, parity error, etc.) the indicator will correspondingly go out.

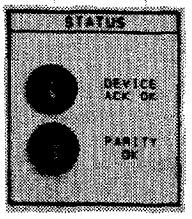
C6. DATA TRANSFER Section. There are three LEDs in the DATA TRANSFER section marked 8, 16 and 32 BITS. These three LEDs are not used and should, therefore, be ignored for all tests. The F ADDRESS MATCH indicator is used in conjunction with the MONITOR mode to indicate a data transfer action associated with a monitored address. The F ADDRESS MATCH indicator is not affected in the SIMULATE mode. The lighting of the F ADDRESS MATCH indicator tells the user that a data transfer was monitored and that the monitored data is displayed on the F ADDRESS MATCH indicator. The F ADDRESS MATCH indicator is latched such that once it is turned on, it remains on until extinguished by setting the MONITOR/SIMULATE switch in the RESET position.

The turn-on of the F ADDRESS MATCH indicator is based on the detection of a match between the address on the digital I/O bus and the address selected on the DTM I/O ADDRESS thumbwheels plus the DATA SELECT switch.

The use of address lines OBAPFO-7 or OBAPRO-7 for data transfer is indicated by the turn-on of the F ADDRESS MATCH indicator.



5. OPERATING CONTROLS (continued)



C7. STATUS Indicators. The status of a MONITOR or SIMULATE data transfer is indicated by two LEDs (DEVICE ACK OK and PARITY OK). These indicators are normally turned on. However, if an error is detected, the appropriate indicator will extinguish. The functions of the two indicators are tabulated as follows:

ERROR INDICATION	APPLICABLE TEST MODE	ERROR CONDITION CAUSE
DEVICE ACK OK	SIMULATE	1. No response from addressed I/O card 2. Digital I/O bus is busy 3. No hold acknowledge
PARITY OK	SIMULATE/ MONITOR	1. Output card has detected a parity error on a SIMULATE-OUTPUT data transfer. 2. The DTM has detected a parity error as a result of a SIMULATE INPUT data transfer from an input card. The erroneous data is displayed on the DATA indicators. 3. The DTM has detected a parity error as a result of monitoring a PMC-to-I/O card data transfer. The erroneous data is displayed on the DATA indicators.

6. DTM CHECKOUT

The DTM checkout is divided into three sections which correspond to tests of the following functions:

A. POWER ON/OFF
B. DIRECT I/O
C. DIGITAL I/O

The following check out procedures are functional checks of DTM operation which do not require the DTM to be dismantled.

NOTE

To check out a DTM, a known serviceable system is required. For the Direct I/O tests, the user must have an electronics enclosure which includes reference manipulation, regulator, firing circuit and a high voltage board. For a Digital I/O test, an enclosure containing a set of functioning I/O cards plus slave station cards is required.

Certain tests specify continuity between a given test point and a certain pin on the maintenance test connector. These pins are very malleable and can be easily bent if care is not taken.



A. POWER ON/OFF. Ensure DTM is properly connected, then set the following POWER ON permissive switches to the settings indicated below:

(1) DIRECT TEST MODE [OFF]
(2) M-CONTACT [OPEN]
(3) LOGIC INPUT ENABLE [OFF]
(4) I/O TEST ENABLE [OFF]
(5) POWER OFF/ON SWITCH [ON]

The DTM power-on condition is indicated by a red LED indicator and the digital voltmeter coming on. If DTM power does not come on, verify that the following conditions do exist:

6. DTM CHECKOUT (continued)

1. The designated switches are in the positions specified above.
2. AC power connection is good.
3. Maintenance test connection is good.
4. Check that front panel AC fuse is good.
5. Verify that P15V and ACOM power exists at electronics enclosure.
6. Inspect maintenance test connector at both ends for bent or damaged pins.
7. Test for a misoriented POWER ON permissive switch (i.e., switch 180° out of position).
8. With the maintenance test cable plug assembly disconnected from the DTM and electronics enclosure, check the cable for the following pin-to-pin continuity:

Block Pin	Signal
A-38	P15
A-39	ACOM
C-31	DCOM
C-33	P5

If steps 1 through 8, above, do not reveal the problems, the failure is probably internal to the DTM and the unit should be replaced.

B. DIRECT I/O. With the DTM properly connected and turned on, check the following control functions using the digital voltmeter (DVM) on the DTM front panel. If the DVM appears to be malfunctioning, an external voltmeter, connected to the appropriate test jacks, can be used.

1. Check REF 1 and REF 2 test references by measuring the test reference output. With the TEST REFERENCE ADJUST knob set fully counterclockwise (CCW) the output voltage (between ACOM and REF 1 and REF 2 output jacks) should be approximately 0.0 volts.

   (a) With the TEST REFERENCE ADJUST knob set fully clockwise (cw) the output voltage should be equal to or slightly greater than $\pm$10.0 volts DC. The voltage polarity should correspond to the NEG (-)/POS (+) switch position.

   (b) Reverse the NEG (-)/POS (+) switch setting to the opposite position. The output polarity should reverse while the voltage magnitude remains approximately as it was.

2. Check the REF 1 STEP function by measuring the REF 1 output voltage. Set the REF 1 STEP adjustment knob fully counterclockwise and set the REF 1 STEP OFF/ON switch to OFF position. Monitoring the digital voltmeter on the DTM, adjust the REF 1 output voltage to +10.0 volts.

6. DTM CHECKOUT (continued)

Set the REF 1 STEP ON/OFF switch to ON position. The voltage monitored on the digital voltmeter should not change.

Rotate the REF 1 STEP adjustment knob clockwise and verify that the monitored voltage increases in amplitude. When the REF 1 STEP adjustment knob is turned fully clockwise, the REF 1 output voltage should be equal to or greater than 11.0 volts.

Set the REF 1 STEP ON/OFF switch to OFF and verify that the REF 1 output voltage goes back to +10.0 volts.

Repeat the above test of the REF 1 output using -10.0 volts. A -1.0 volt step to a REF 1 output of -11.0 volts should be observed.

3. The SUPPRESSION TEST function can be checked by simulating a suppression test. This is accomplished by setting the REF 2 adjustment knob to its maximum positive value and setting the TEST CHARGE switch in the CHARGE position. Read the suppression voltage on the DTM digital voltmeter. This voltage should be equal to, or greater than +10.0 volts.

With the SUPPRESSION TEST ON/OFF switch set to the OFF position, set the TEST/CHARGE switch in the TEST position. The measured voltage should decay to approximately 0.0 volt.

(a) An alternate method of troubleshooting the suppression test reference is accomplished by verifying that the test pulse generated by an electronic module will trip the firing circuit suppression circuit. When the suppression circuit is tripped, the suppression indicator (LED) on the firing circuit card will be extinguished. A toggle switch on the firing circuit card is provided to re-enable the suppression circuit and illuminate the LED indicator. To check the module suppression test pulse, proceed as follows:

(1) Set the REF 2 output to its maximum positive value (greater than +11.5 VDC).

(2) Reset the firing circuit suppression circuit if it has been tripped.

(3) If the DTM SUPPRESSION TEST circuit is not internally wired to the firing circuit suppression circuit (a requisition variable):

6. DTM CHECKOUT (continued)

(a) shut off power to the high voltage board,

(b) jumper the SUPPRESSION TEST output to the positive (+) side of the burden resistor. An ACOM jumper is not necessary since the negative side of the burden resistor is common with the test circuit common, and

(c) reapply power to the high voltage board.

WARNING

EXTREME CARE SHOULD BE EXERCISED WHEN WORKING ON THE HIGH VOLTAGE BOARD DUE TO THE EXISTENCE OF HIGH VOLTAGE.

(4) Initiate the suppression test by depressing the suppression toggle switch on the DTM. The firing circuit should suppress. If the suppression LED indicator remains illuminated, the suppression test circuit of the DTM is suspect.

4. The digital voltmeter may be tested by setting the TEST POINT SELECT switch to EXT/GROUP B position and injecting a calibrated voltage through the EXTERNAL INPUT TO METER test jacks. The test voltage source can be a REF 1 to REF 2 test reference calibrated with an external voltmeter.

5. Checkout of the DIRECT TEST MODE requires that the user select a test mode position and verify the appropriate response at the drive electronics regulator card, firing circuit card and reference manipulation card LED indicator. If the electronics enclosure backplane is accessible, the user can verify proper operation by use of the truth table on the following page.

6. DTM CHECKOUT (continued)

SIGNAL (BLOCK-PIN) / TEST MODE POSITION	OSUA (C-37)	ORLT (C-39)	OSUS (C-38)	ORLA (C-40)	ORLR (C-21)
OFF	1	1	1	1	1
SUICIDE ALL	0	1	1	1	1
RELEASE FIRING CIRCUIT SUICIDE	1	0	1	1	1
RELEASE FIRING CIRCUIT AND REGULATOR IN LOOP SUICIDES	1	1	0	1	1
RELEASE FIRING CIRCUIT AND REGULATOR SUICIDE	1	1	1	0	1
RELEASE ALL	1	1	1	0	0

6. Check the TEST POINT SELECT switches by reading the output of one of the two rotary switches while using the GROUP A/GROUP B select switch. The response can be checked on the DTM front panel digital voltmeter. Read the power bus voltage, P15, N15, P28/2 and P5. (Note that the P5 voltage will vary by the voltage differential between ACOM and DCOM.)

   a. Read REF 1, REF 2 and SUPP test reference.

   b. Inject a known reference into the EXT input test jacks (e.g. from the REF 1 test reference).

   c. The remaining test points (points 1-8 on GROUP A and points 1-8 on GROUP B) can be best checked by continuity tests between the GROUP A or GROUP B test jack and the unconnected end of the maintenance test cable plug. Continuity should be measured per the following table.

TEST POINT SELECTOR A POSITION	CONTINUITY GROUP A TEST JACK TO
1	A-2
2	A-9
3	A-15
4	A-14
5	A-3
6	A-4
7	A-5
8	A-6

TEST POINT SELECTOR B POSITION	CONTINUITY GROUP B TEST JACK TO
1	A-10
2	A-13
3	A-17
4	A-20
5	A-7
6	A-12
7	A-16
8	A-22

6. DTM CHECKOUT (continued)

7. The DIRECT LOGIC TEST function requires that the user perform the following actions:

   a. Check LOGIC OUTPUT by correlating each output LED indicator with its respective signal connection at the electronics enclosure. (Note: 0.0V = indicator extinguished and 5.0V = indicator lit.) The logic output signals can be located by reference to Table 1.

   b. Test the LOGIC INPUT ENABLE switch by moving the LOGIC INPUT switches while checking for corresponding actions at the electronics enclosure. (Note: 5.0V = "1" and 0.0V = "0".) The logic input signals can be located by reference to Table 2.

   c. Verify that all the LOGIC INPUT signals do not go to "0" (0.0V) in the "1" switch position if the LOGIC INPUT ENABLE switch is off.

Table 1 - Location of Logic Output Points

LOGIC OUTPUT POINT NO.	MAINT. TEST CONN (BLOCK-PIN)
1	A-1
2	A-11
3	A-18
4	A-21
5	A-28
6	A-36
7	A-37
8	C-15
9	C-17
10	C-19

Table 2 - Location of Logic Input Points

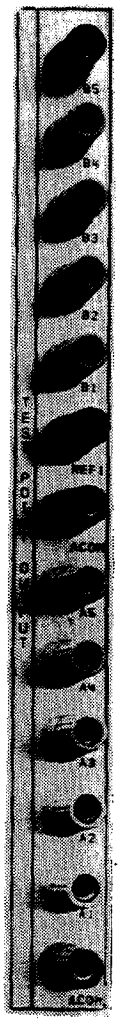
LOGIC INPUT POINT NO.	MAINT. TEST CONN (BLOCK-PIN)
1	C-14
2	C-16
3	C-18
4	C-20
5	C-22
6	C-24
7	C-25
8	C-26
9	C-27
10	C-28

6. DTM CHECKOUT (continued)



8. SIGNAL ACCESS test jacks can be tested by checking continuity between the SIGNAL ACCESS test jacks and corresponding pins on the maintenance test connector plug. Points of continuity are shown in the following table:

SIGNAL ACCESS Test Jack	Maintenance Test Connector (Block-Pin)
SA1	A-19
SA2	A-25
SA3	A-23
SA4	A-27
SA5	A-29
SA6	A-33
SA7	A-35
SA8	A-24
SA9	A-26
SA10	A-30
SA11	A-32
SA12	A-34

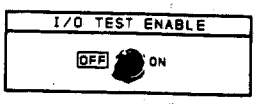


9. The TEST POINT OUTPUT test jacks can be checked out by measuring for continuity between the test jacks and the TEST POINT SELECT GROUP A or GROUP B test jacks. The test requires that the user check for 10 K-ohms of continuity between each of the positions shown in the following table. (This test is accomplished under the presumption that the TEST POINT SELECT switches are functioning properly):

TEST POINT OUTPUT Test Jack	TEST POINT SELECT Switch	TEST POINT SELECT Switch Positions
A1	A	1
A2	A	2
A3	A	3
A4	A	4
A5	A	5
B1	B	1
B2	B	2
B3	B	3
B4	B	4

6. DTM CHECKOUT (continued)

C. DIGITAL I/O. The following test requires that the DTM be connected to, and provided power by a functioning digital I/O electronics enclosure. The enclosure should have an 8 and 16-bit input card and an 8 and 16-bit output card. The I/O addresses of these cards must be known. The knowledge of an I/O address to which no I/O card is assigned is not a valid I/O address.



1. The I/O TEST ENABLE switch, when in the OFF position, isolates the DTM from the digital I/O bus and turns off the DATA LED indicators. Switching the I/O TEST ENABLE on will cause a random pattern to appear on the DATA indicators.

2. Check for a "burned-out" DATA LED indicator by turning on all the DATA indicators as follows:

   a. Turn on DATA indicators 0 thru 7 by:

      (1) setting the DATA SELECT switch to the LOWER BYTE position, and

      (2) selecting the MONITOR mode and then depressing and releasing the MONITOR/SIMULATE switch to the RESET position.

   b. Turn on DATA indicators 8 thru F using the same procedure as above, except set the DATA SELECT switch to the UPPER BYTE.

3. Verify that the MONITOR/SIMULATE modes can be enabled by placing the I/O TEST MODE switches in the positions listed below and checking for the MONITOR/SIMULATE STATUS RUN indicator being turned on.

   a. MODE switch set to MONITOR
   b. SINGLE/CONTINUOUS switch set to CONTINUOUS (NO ERROR STOP).
   c. MONITOR/SIMULATE switch set to INITIATE, then released.

   If RUN indicator does not come on, ensure that a reset condition is not present at the digital I/O bus. This is accomplished by placing the appropriate digital bus interface card (e.g. HDBA or HPBA card) in the MAINTENANCE or ON mode.

6. DTM CHECKOUT (continued)

4. Check the MONITOR mode operation as follows:

   a. Place the appropriate digital interface card into the "ON" mode.

   b. Select an I/O address to which the PMC periodically selects.

   c. Select the I/O TEST MODE INPUT/OUTPUT switch to the corresponding type of card that the PMC is referencing.

   d. Depress and release the MONITOR/SIMULATE switch to the RESET position. This should turn off any DATA TRANSFER LEDs that may be lit.

   e. Depress and release the MONITOR/SIMULATE switch to the INITIATE position. The RUN indicator should be lit and the DATA transfer LED(s) should turn on. Depending on the type of bus, either all of the DATA transfer LEDs should be lit (R-bus connection) or just the F ADDRESS MATCH LED should be lit (F-bus connection).

   f. Verify that the monitored data corresponds to the INPUT or OUTPUT data per the type I/O card being monitored.

   g. Depressing and releasing the MONITOR/SIMULATE switch to the RESET position should extinguish the DATA transfer and MONITOR/SIMULATE STATUS RUN LEDs.

   h. The test should be repeated using the opposite type of I/O card, e.g. switch the I/O TEST MODE INPUT/OUTPUT switch to OUTPUT if the previous test was accomplished by use of the INPUT switch position.

   j. Repeat steps a thru g but use the I/O TEST MODE SINGLE switch position. The user should see the RUN indicator blink on, then extinguish and the DATA transfer LEDs should come on and remain on.

5. SIMULATE MODE operation may be tested by performing a data transfer to a known working input card. The test sequence should follow the steps listed in Paragraph 7B. The I/O mode condition should be CONTINUOUS (ERROR STOP) and the digital interface card should be in the MAINTENANCE mode. The user should observe the following conditions:

   a. The MONITOR/SIMULATE STATUS RUN indicator should come on.
   b. Both STATUS indicators (DEVICE ACK OK and PARITY OK) should be lit.
   c. The DATA indicators should reflect the state of the input card.

6. DTM CHECKOUT (continued)

Execute the test for 8 and 16-bit input cards. Select the appropriate DATA SELECT switch position per the input card type and address.

6. Test the SIMULATE-OUTPUT mode operation by performing a data transfer from the SIMULATE DATA switches to a known functional output card. The test sequence should follow the steps listed in Section 7, paragraph A. The I/O test mode condition should be CONTINUOUS (ERROR STOP) and the appropriate digital bus interface should be in the MAINTENANCE mode. The user should observe the following conditions:

   a. The MONITOR/SIMULATE STATUS RUN indicator should come on.

   b. Both STATUS indicators (DEVICE ACK OK and PARITY OK) should be lit.

   c. The user should be able to:

      (1) change the output data via the SIMULATE DATA switches while the mode is active and observe the change at the output card.

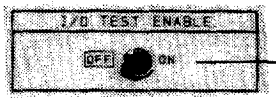
      (2) ensure that the DATA indicators should follow the switch position (i.e., a SIMULATE DATA switch up (logic "1") should cause the corresponding DATA indicator to be lit, and conversely the switch down (logic "0") should extinguish the DATA indicator.

7. By selecting an invalid I/O address, a SIMULATE DATA transfer should cause the DEVICE ACK OK indicator to be extinguished when the test condition is CONTINUOUS (ERROR STOP). Selecting the test mode condition CONTINUOUS (NO ERROR STOP) should not extinguish the DEVICE ACK OK indicator when a SIMULATE data transfer operation is executed.

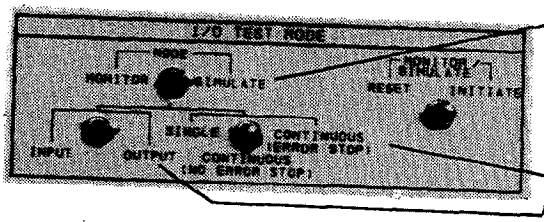
7. DIGITAL I/O OPERATING PROCEDURE

(DTM is the active device)

A. SIMULATE OUTPUT - causes a DTM to output card data transfer.



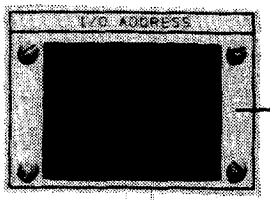
Enable I/O Test
NOTE: The slave station controller card (DS3800HLSD) must be in the maintenance mode.



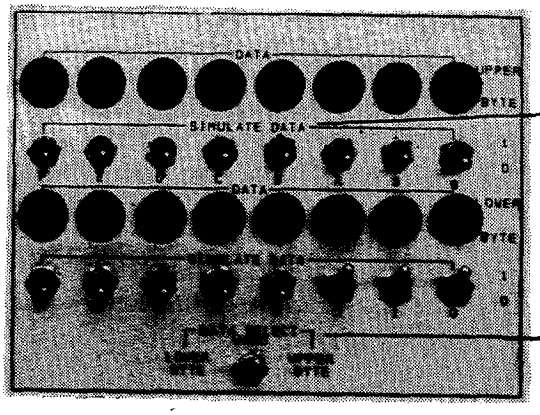
Select SIMULATE Mode

Select OUTPUT Mode

Choose Test Conditions

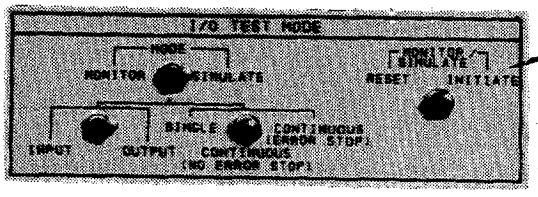


Define I/O Address



Choose DATA SELECT Mode

Set up SIMULATE DATA switches

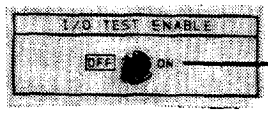


Start test - Depress and release MONITOR/SIMULATE - RESET/INITIATE switch to INITIATE position.

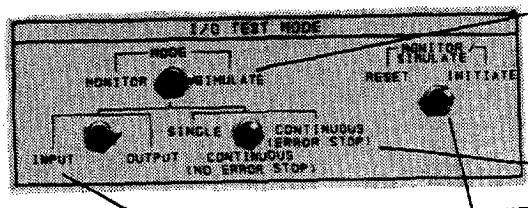
7. DIGITAL I/O OPERATING PROCEDURE (continued)

B. SIMULATE INPUT - causes an input card-to-DTM data transfer.

(DTM is the active device)



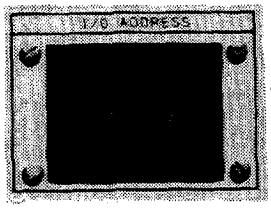
Enable I/O TEST



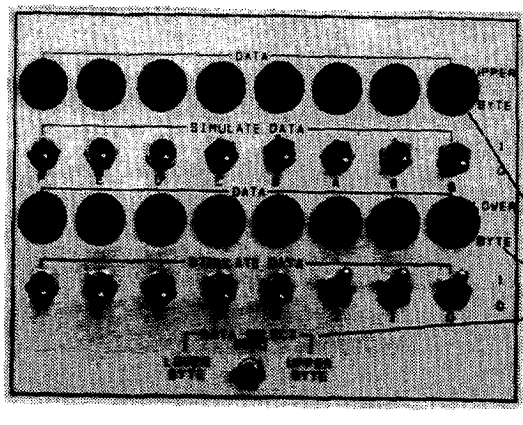
Select SIMULATE mode

Select INPUT mode

Choose test condition



Define I/O address



Choose DATA SELECT WORD.

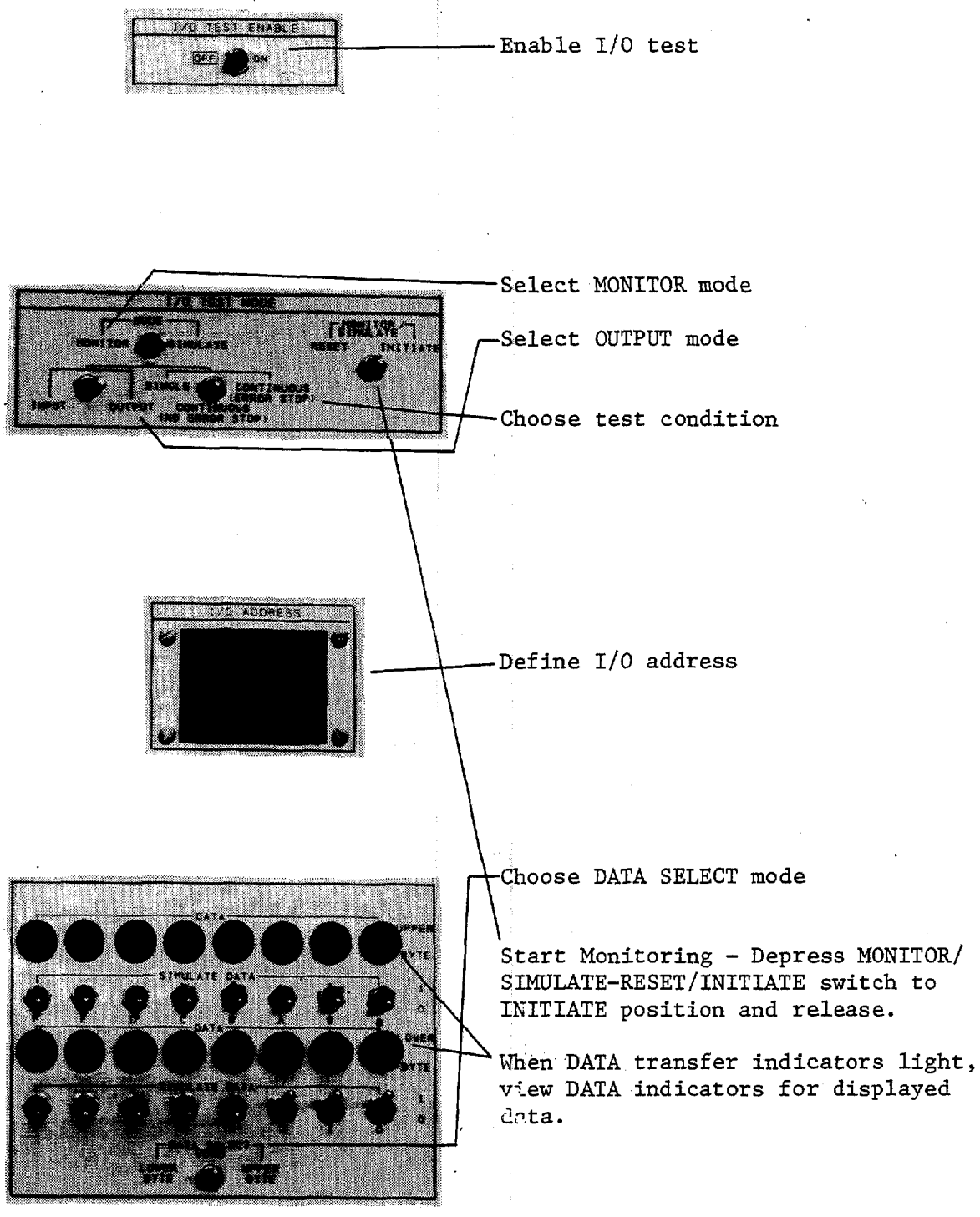
Start simulate input - Depress and release MONITOR/SIMULATE RESET/ INITIATE switch to INITIATE position.

Monitor DATA indicators for input data.

7. DIGITAL I/O OPERATING PROCEDURE (continued)

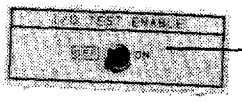
C. MONITOR OUTPUT - Monitor PMC-to-output card data transfer.

(DTM is the passive device.)

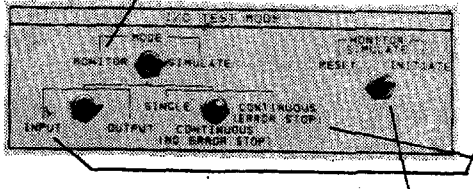


7. DIGITAL I/O OPERATING PROCEDURE (continued)

D. MONITOR INPUT - Monitor input card-to-PMC data transfers.



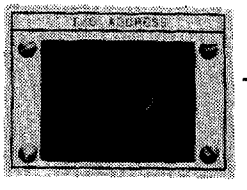
Enable I/O Test



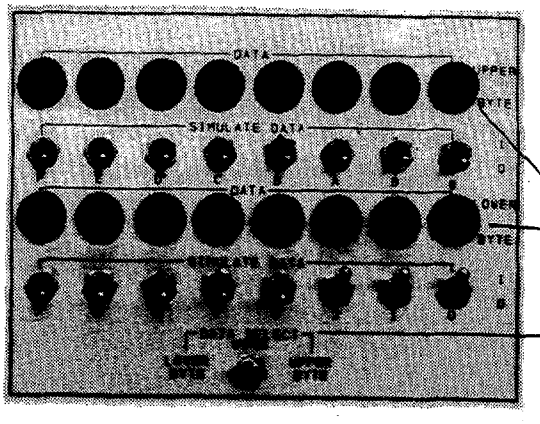
Select MONITOR mode

Select INPUT mode

Choose test condition



Define I/O address



Choose DATA SELECT WORD

Start monitoring - Depress MONITOR/ SIMULATE RESET/INITIATE switch to INITIATE position, then release.

When DATA transfer indicators light, view DATA indicators for displayed data.

8. DTM MAINTENANCE INSTRUCTIONS

A. Periodic Maintenance. The only periodic maintenance requirement is that of a digital voltmeter (DVM) calibration check. This should be performed once each year. To check and adjust the digital voltmeter, proceed as follows:

(1) Apply a known calibration voltage (+19.99 VDC or less) to the EXTERNAL INPUT TO METER terminals, turn the GROUP B TEST POINT SELECT switch to EXT. position and check DVM calibration. If an adjustment is required, perform steps (2) through (6).

(2) Remove calibration voltage from DVM input terminals.

(3) Using fingers, slide the DVM red lens forward and observe a small opening in the bottom-center of the lens. Carefully insert any convenient tool into the hole and lift the lens away, thereby exposing the meter works.

(4) Re-apply the calibrating voltage as in step (1) above. On the lower right-hand side of the meter works is a DVM read-out adjustment screw. Adjust the DVM reading to coincide with the calibrating voltage.

(5) Disconnect the calibrating voltage.

(6) Re-insert the DVM read lens.

B. Module Operating Check. The following procedure allows a rapid and simple check to verify proper operation of the DTM.

(1) Perform a power distribution check as follows:

(a) Plug in DTM AC cord and signal cable.

(b) Set the following switches to positions indicated:

SWITCH	POSITION
DIRECT TEST MODE Selector	OFF
M CONTACT	OPEN
LOGIC INPUT ENABLE	OFF
I/O TEST ENABLE	OFF
POWER	ON

(c) Verify that the DTM POWER LED (red) comes on and that the digital voltmeter display appears.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(2) Perform a permissive interlock check as follows:

(a) Set DTM POWER switch to OFF position.

(b) Set any one of the following switches to positions indicated below:

SWITCH	POSITION
DIRECT TEST MODE Selector	RELEASE ALL SUICIDES
M CONTACT	CLOSED
LOGIC INPUT ENABLE	LOGIC INPUT ENABLE
I/O TEST ENABLE	ON

(c) Set POWER switch to ON position and verify that the DTM does not come on.

(3) Perform a bus voltage check as follows:

(a) Monitoring digital voltmeter, set TEST POINT SELECT switch(es) to P5, P15, N15 and P28/2 positions and verify the presence of specified voltages.

(4) Perform TEST REFERENCE voltage check using the digital voltmeter. Using TEST REFERENCE 1 circuit, verify the operation of the REF 1 STEP circuit.

NOTE

The TEST REFERENCE OFF/ON Switch has no affect on DVM readings.

(5) Perform a SUPPRESSION TEST circuit check as follows:

NOTE

The following procedure is performed without a test load.

(a) Set TEST POINT SELECT switch to SUPP position.

(b) Monitoring digital voltmeter, charge SUPPRESSION TEST circuit to voltage level set by TEST REFERENCE 2 circuit.

(c) Monitoring digital voltmeter, set SUPPRESSION TEST switch to TEST position and verify that an instantaneous voltage discharge occurs.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(6) Verify digital I/O monitor mode operation as follows:

(a) Set the following switches to indicated positions:

SWITCH	POSITION
I/O TEST ENABLE	ON
I/O ADDRESS thumbwheel	Any Address
INPUT/OUTPUT	INPUT
MODE	MONITOR
SINGLE/CONTINUOUS	CONTINUOUS (NO ERROR STOP)

(b) Set MONITOR/SIMULATE switch to INITIATE position and verify that the MONITOR/SIMULATE STATUS LED (RUN) comes on and remains on.

(7) Verify digital I/O simulate mode operation as follows:

(a) Set the control signal highway slave station card to the "maintenance" mode.

(b) Set the following switches to the positions indicated:

SWITCH	POSITION
I/O TEST ENABLE	ON
I/O ADDRESS thumbwheel	Any Address
INPUT/OUTPUT	INPUT
MODE	SIMULATE
SINGLE/CONTINUOUS	CONTINUOUS (NO ERROR STOP)

(c) Set MONITOR/SIMULATE switch to INITIATE position and verify that the MONITOR/SIMULATE STATUS LED (RUN) comes on and remains on.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

C. DTM Troubleshooting (Analog Circuitry). Refer to Appendix for DTM elementary diagram and backplane pin assignment data. Figures 10, 11 and 12 show DTM interior wiring and hardware details.

NOTE

The following tests are to be performed with the DTM removed from its case, thereby allowing access to the NDTA card, internal connections and wiring. The DTM must be connected to an operating system (via the maintenance test connector) and AC power must be applied (via the DTM AC input cable).

WARNING

WHILE THE DTM IS REMOVED FROM ITS CASE, LIVE VOLTAGES ARE PRESENT ON EXPOSED WIRING AND HARDWARE. THEREFORE, PRECAUTIONS SHOULD BE TAKEN TO AVOID A SHOCK HAZARD.

(1) SYMPTOM: DTM power will not come on as indicated by POWER LED.

(a) Check for secure AC power cord and DTM signal cable connections.

(b) Check AC power fuse on DTM front panel.

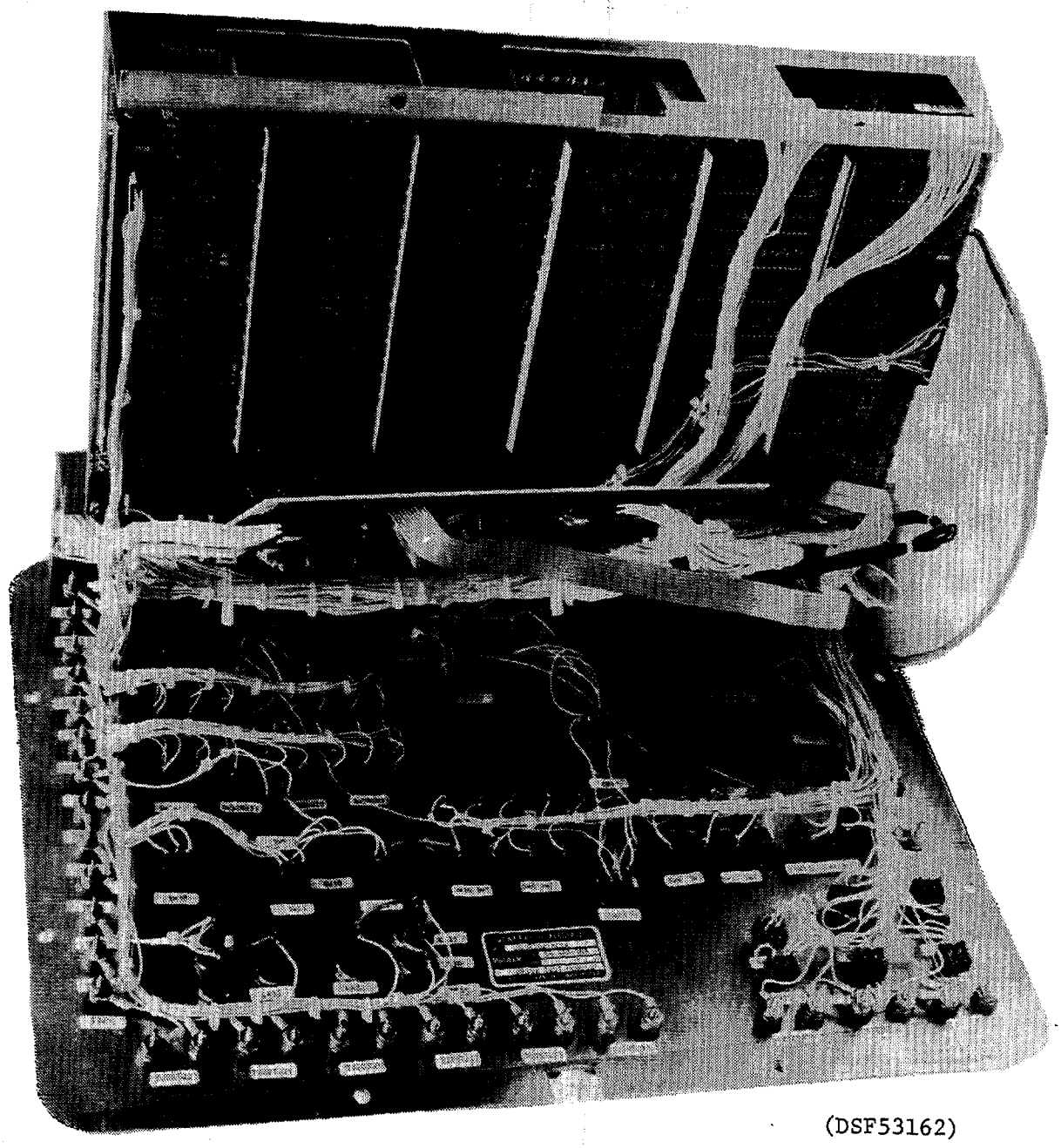
(c) Using a DC voltmeter, check P15 and N15 outputs at GROUP A and GROUP B test jacks.

(d) Verify that the following switches are set to indicated positions:

SWITCH	POSITION
DIRECT TEST MODE	OFF
M CONTACT	OPEN
LOGIC INPUT ENABLE	OFF
I/O TEST ENABLE	OFF

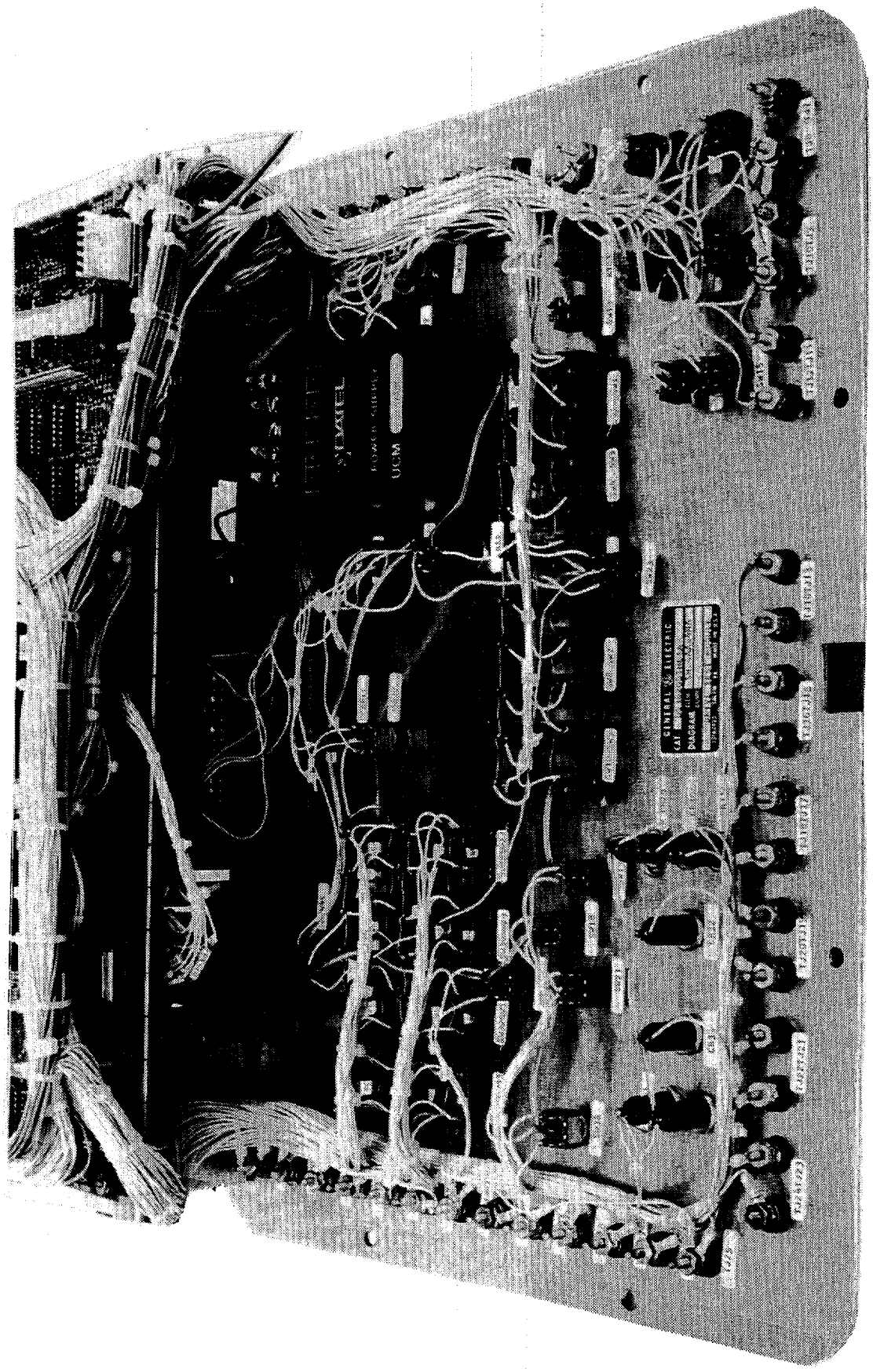
(e) Check for presence of internal P5 power supply output.

(f) Check integrity of connection between DTM internal P5 power supply and NDTA board.



(DSF53162)

Figure 10. DTM Interior View (NDTA Board Folded Outward Exposing HDTA Board)



(DSF53161

Figure 11. Rear View of DTM Front Panel

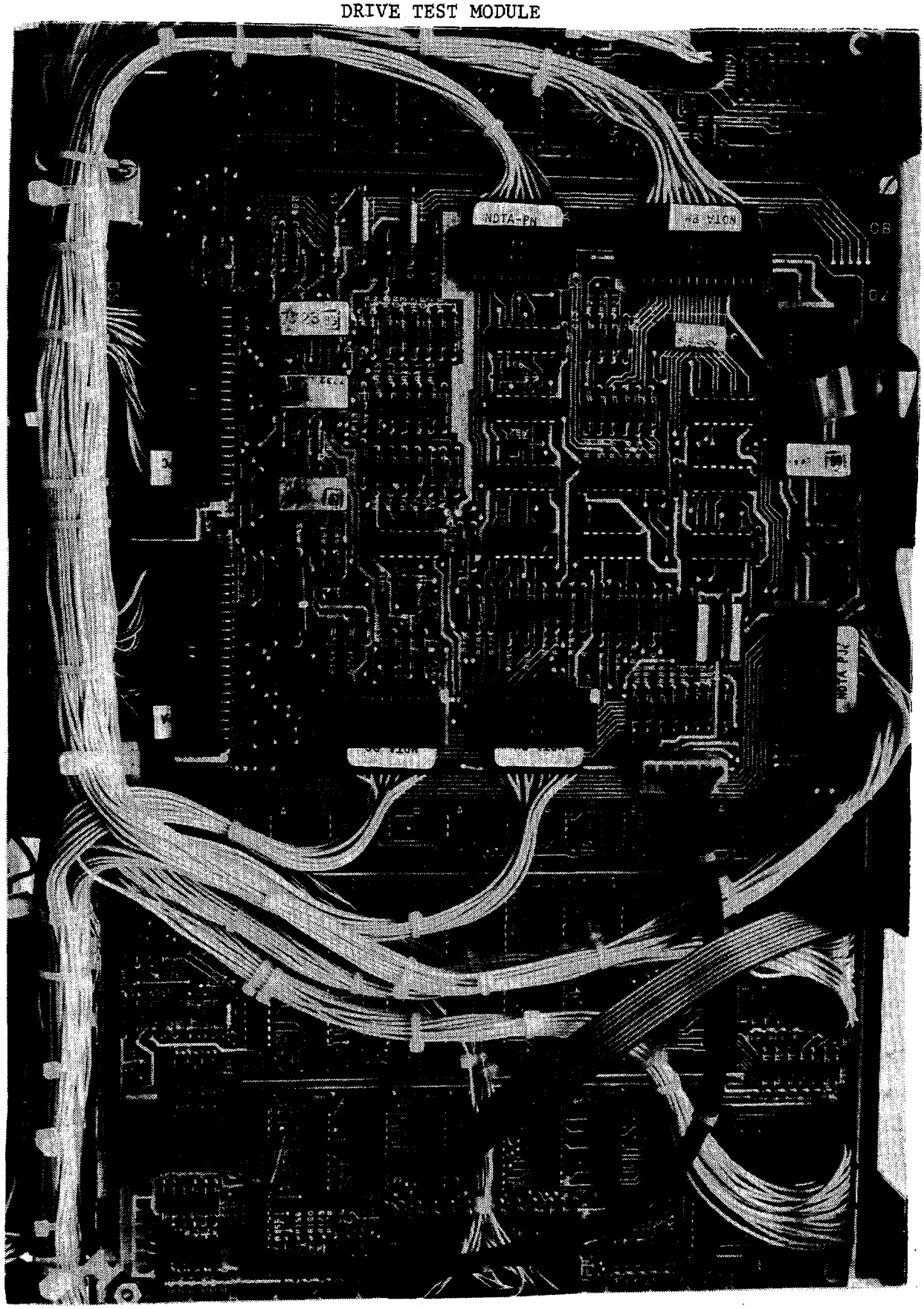


Figure 12. NDTA Board Detail (DSF53163)

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(g) Check maintenance test connector for loose, unseated, or bent pins.

(h) If steps (a) through (g) do not provide a solution to the problem, remove and replace the NDTA board.

(2) SYMPTOM: DTM will not power off.

(a) Remove plug PN from NDTA board; check for short circuit to ground between internal power supply and point PN-9. If a short circuit is found, faulty wiring or a shorted switch is the cause of the problem.

(b) Check POWER switch for a short to ground.

(c) If items (a) or (b) do not reveal the solution to the problem, replace the NDTA board.

(3) SYMPTOM: Digital voltmeter does not come on when POWER LED comes on.

(a) Check for presence of ≈5.0 VDC to meter between PZ-15 and PZ-S. (See PZ plug pin markings on following page.)

(b) If step (a) does not reveal the solution to the problem, replace the digital voltmeter.

(4) SYMPTOM: DTM front panel fuse blows when AC power cord is plugged in.

(a) Replace internal P5 power supply.

(5) SYMPTOM: Signals selected by TEST POINT SELECT switches are not displayed on digital voltmeter.

(a) Check for shorted or broken wires at rear of TEST POINT SELECT GROUP A and GROUP B selector switches.

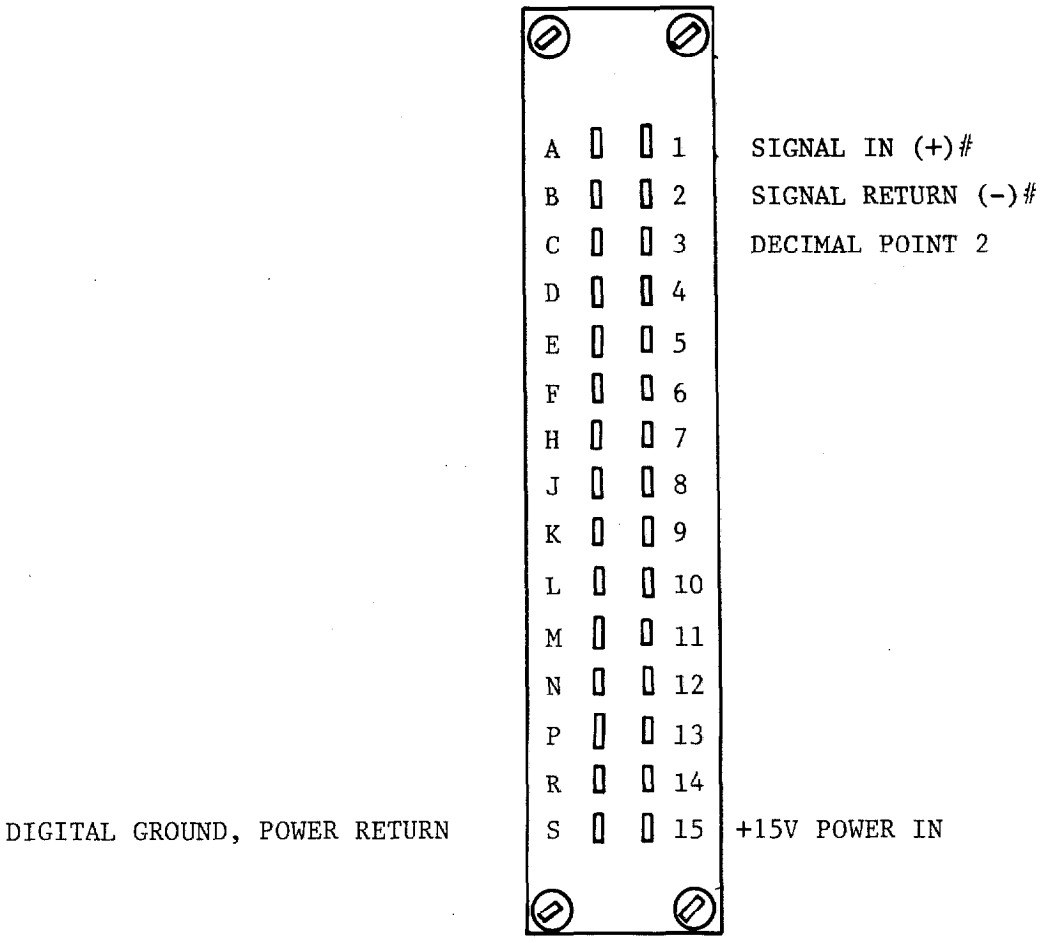
(b) Check for presence of test signal at GROUP A and/or GROUP B test jacks.

1. Signal present at test jack:

Cause: Faulty wiring to digital voltmeter or faulty digital voltmeter.

2. Signal not present at test jack:

Cause: Misaligned GROUP A or GROUP B selector knob or broken/misconnected wiring between DTM and connected system.



DIGITAL METER PLUG (PZ) PIN MARKINGS

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(6) Direct test mode malfunction.

(a) Check for broken or shorted wires at rear of DIRECT TEST MODE selector switch.

(b) Using a DC voltmeter, verify the presence of logic signals on the system maintenance test connector backplane as shown in Table 1. The backplane pin layout of the maintenance test connector is shown in Figure 13.

CAUTION

Care should be taken when attaching meter probes to connector pins to prevent damage to pins or shorting pins together.

DIRECT TEST MODE SWITCH POSITIONS	MAINT. TEST CONN BACK PLANE PINS				
	C-37	C-39	C-38	C-40	C-21
OFF	1	1	1	1	1
SUICIDE ALL	0	1	1	1	1
RELEASE FIRING CIRCUIT	1	0	1	1	1
RELEASE F.C. AND REG. IN LOOP	1	1	0	1	1
RELEASE F.C. AND REG.	1	1	1	0	1
RELEASE ALL	1	1	1	0	0

Table 1. DIRECT TEST MODE Logic Matrix

8. DTM MAINTENANCE INSTRUCTIONS (continued)

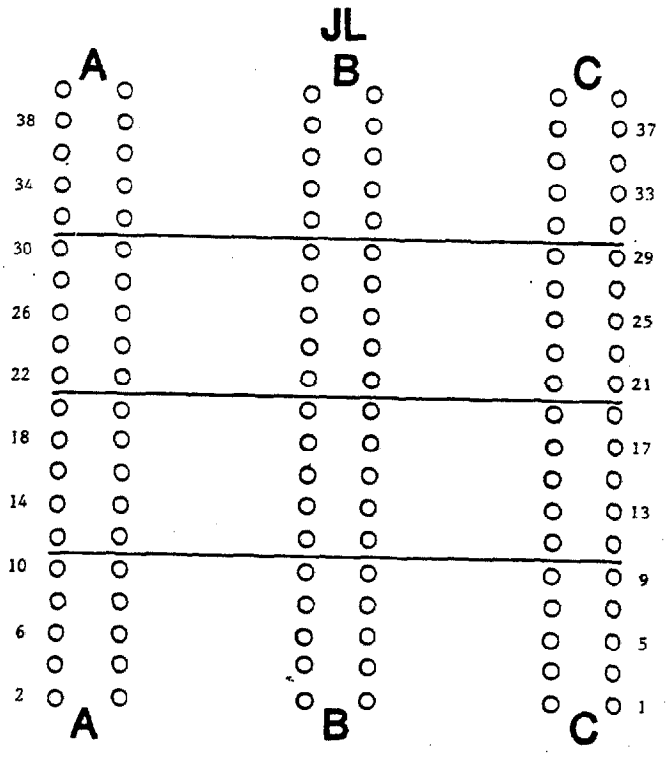


Figure 13. Backplane Pin Layout of Maintenance Test Connector.

(7) SYMPTOM: DIRECT TEST MODE logic signals present on pins of maintenance test connector, but do not correspond with those in Table 1.

(a) Improperly wired maintenance test connector.

(b) DIRECT TEST MODE selector knob improperly aligned.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(8) SYMPTOM: DIRECT TEST MODE signals not present on pins of maintenance test connector.

(a) Power common not present at center pin of DIRECT TEST MODE selector switch.

(b) Loose plug or poor connection at PN-JN or PC-JC on NDTA board.

(c) Unseated pin on plug PN or PC.

(d) Faulty DIRECT TEST MODE selector switch.

(e) Faulty NDTA board.

(9) SYMPTOM: Signals present which correspond to those in Table 1 but suicide control causes M-contactor to trip.

(a) Wrong type of DIRECT TEST MODE switch installed. (Switch should be a shorting - type - "Make before break".)

(10) M CONTACT toggle switch inoperative.

(a) Inspect switch for shorted or broken wiring.

(b) Check for loose plug or plug-receptacle connection at PN-JN or PC-JC on NDTA board.

(c) Check for unseated or bent pin at maintenance test plug.

(d) Check for unseated pin in plug PN or PC.

(e) Replace NDTA board.

(11) M CONTACT LED inoperative.

(a) Burned out LED.

(b) Check for unseated or bent pin at maintenance test plug.

(c) Check for loose plug-receptacle connection at PM-JM or PC-JC on NDTA board.

(d) Replace NDTA board.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(12) DIRECT LOGIC TEST - LOGIC INPUT switch(es) inoperative.

(a) Inspect switch(es) for shorted or broken wiring.

(b) Check for loose plug - receptacle connection at JM-PM, JA-PA, and JC-PC.

(c) Check for unseated pin on maintenance test connector plug.

(d) Check for unseated pin plugs PM, PA and PC.

(e) Replace NDTA board.

(13) SYMPTOM: DIRECT LOGIC TEST - LOGIC OUTPUT LED will not come on.

(a) Check for burned-out LED.

(b) Check for unseated or bent pin at maintenance test connector.

(c) Replace NDTA board.

(14) TEST REFERENCE function inoperative.

(a) Check wiring - side of TEST REFERENCE panel for broken wire or bent pin at respective TEST REFERENCE potentiometer.

(b) Check for loose PH-JH connection on NDTA board or a bent pin at plug PH.

(c) With power on, verify the presence of P15 or N15 between pin 1 and pin 3 of potentiometer by use of an external voltmeter. If no voltage is present, proceed as follows:

1. Check for loose PH-JH connection at NDTA board or for an unseated pin at PH.

2. Verify the presence of P15 and N15 voltage at NDTA board.

3. Check for broken wire or a shorted pin on TEST REFERENCE ON/OFF.

4. Replace NDTA board.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(15) REF 1 or REF 2 output provides only one polarity output.

(a) Perform procedural steps given in (14) previously.

(b) Check for broken wire or bent tab at respective TEST REFERENCE NEG (-) POS (+) switch.

(c) If external voltmeter indicates correct current and voltage polarity but DTM digital voltmeter does not, verify that P5 at NDTA board is 5.0V $\pm$ 5%.

(16) TEST REFERENCE 1 STEP Circuit inoperative.

(a) Check for broken wire or bent pin at STEP REFERENCE potentiometer and STEP ON OFF switch.

(b) Check for loose PH-JH connection at NDTA board or a bent pin at plug PH.

(c) Verify that the step voltage is being developed at STEP REFERENCE potentiometer (pin 1 to pin 2). If not, the potentiometer must be replaced.

(d) Replace NDTA board.

(17) Digital voltmeter inoperative.

(a) Check for loose PZ edge connector at meter.

(b) Check for a broken or shorted wire at PZ edge connector.

(c) Verify the presence of 5.0V $\pm$ 5% between PZ-15 and PZ-S. If not, DTM internal power supply is probably faulty. Otherwise, the digital voltmeter is faulty and should be replaced.

(18) SYMPTOM: Digital voltmeter displays only negative voltages.

(a) Verify the presence of 5.0V $\pm$ 5% between PZ-15 and PZ-S. If not, DTM power supply is probably faulty. Otherwise, the digital voltmeter is faulty.

8. DTM MAINTENANCE INSTRUCTIONS (continued)

D. DTM Troubleshooting (Digital I/O Section). Refer to Appendix for DTM elementary diagram and backplane pin assignment data. Figures 10, 11 and 12 show DTM interior wiring and hardware details.

NOTE

The following tests are to be performed with the DTM removed from its case, thereby allowing access to the NDTA board, internal connections and wiring. The DTM must be connected to an operating system (via the maintenance test connector) and AC power must be applied (via the DTM AC input cable).

WARNING

WHILE THE DTM IS REMOVED FROM ITS CASE, LIVE VOLTAGES ARE PRESENT ON EXPOSED WIRING AND HARDWARE. THEREFORE, PRECAUTIONS SHOULD BE TAKEN TO AVOID A SHOCK HAZARD.

(1) SYMPTOM: Digital I/O section of DTM will not operate.

(a) Check integrity of the following plug-receptacle connections:

PLUG-RECEPTACLE	LOCATION
PN-JN	NDTA board
PV-JV	NDTA board
PV-JV	HDTA board
PR-JR	HDTA board

(b) Check I/O TEST ENABLE switch wiring.

(c) Replace HDTA board.

(2) SYMPTOM: Digital I/O section turns on but monitor or simulate mode circuitry is inoperative.

(a) Set the following I/O TEST MODE switches to indicated positions:

SWITCH	POSITION
MODE	MONITOR
INPUT-OUTPUT	INPUT
SINGLE-CONTINUOUS	CONTINUOUS (NO ERROR STOP)
MONITOR/SIMULATE	INITIATE

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(b) Select any I/O address.

If DEVICE ACK OK LED goes out, replace HDTA card.

Otherwise, an addressing fault is indicated (see 2. a).

8. DTM MAINTENANCE INSTRUCTIONS (continued)

(b) Set control signal highway slave station card in maintenance mode.

1. If DTM RUN LED does not come on:

a A reset condition is present on the digital I/O bus (thus inhibiting DTM operation) or

b The HDTA board is faulty.

2. IF DTM RUN LED comes on:

a An addressing fault is caused by:

- Faulty DTM hex thumbwheel switch, or
- Faulty address strapping at test system (in particular HACA or DACA) or
- Loose or unseated pin(s) on maintenance test plug, or
- Faulty HDTA board.

(3) SYMPTOM: Monitor mode is active, as indicated by RUN LED, but no address match occurs.

(a) No accesses to digital I/O bus are being performed:

1. Test system's control signal highway slave station card is in maintenance mode, thereby inhibiting accesses, or

2. Faulty control signal highway cards, or

3. PMC not transmitting.

(b) Accesses to digital I/O bus are being performed.

1. Addressing fault (See 2. a).

(4) SYMPTOM: Simulate mode is active (as indicated by RUN LED) but DEVICE ACK OK error occurs.

(a) Set the following I/O TEST MODE switches to indicated positions:

SWITCH	POSITION
MODE	SIMULATE
INPUT-OUTPUT	INPUT
SINGLE-CONTINUOUS	CONTINUOUS (NO ERROR STOP)
MONITOR/SIMULATE	INITIATE

9. REPLACEMENT PARTS

Following is a list of the recommended spare parts for the DTM.

NOMENCLATURE	CATALOG NUMBER
CARD (Analog)	DS3800NDTA
CARD (Digital)	DS3800HDTA
CARD (Common)	DS3800XDTA
POWER SUPPLY	218A4894P1
SWITCH (MA-NONE-MA)	218A4861P1
SWITCH (MA-OFF-MA)	218A4861P2
SWITCH (MO-OFF-MO)	218A4861P3
SELECTOR (Make-Before-Break)	218A4884P2
SELECTOR (Break-Before-Make)	218A4829P2
SELECTOR KNOB	218A4811P1
DIGITAL PANEL METER	218A4085P1
LED (Red)	218A4867P1RD
LED (Yellow)	218A4867P1YL
POTENTIOMETER	68A7130P252
POTENTIOMETER KNOB	68A7612P1
TEST JACK (Red)	162A5442P3
TEST JACK (Blue)	162A5442P2
TEST JACK (Black)	162A5442P1
TEST JACK (White)	162A5442P5
FUSE (0.25A "Fast")	573A392P66
CARRYING CASE	218A4853P1
CABLE-UMBILICAL	206B3312602
CABLE, RIBBON (10-Wire)	DS3815RCA1CL018X00
CABLE (NDTA-to-Panel)	68A947268G1L018
CABLE (NDTA-to-Panel)	68A947268G2L028
CABLE (NDTA-to-Panel)	68A947268G3L038
CABLE (NDTA-to-Panel)	68A947268G4L045
CABLE (NDTA-to-Panel)	68A947268G5L035
CABLE (NDTA-to-Panel)	68A947268G6L032
CABLE (NDTA-to-Panel)	68A947268G7L045
CABLE (NDTA-to-Panel)	68A947268G8L006
CABLE (NDTA-to-Panel)	68A947268G9L025
CABLE - AC Power	218A4893P1
CABLE (PWB-PWB)	68A947194G1
THUMBWHEEL Switch	218A4009B2CJ
FUSE HOLDER	573A389P2
RECEPTACLE FOR AC POWER CORD	218A4892P1

10. APPENDIX

This appendix contains maintenance test connector pin assignments and DTM system elementaries, useful for maintenance and troubleshooting purposes.

Maintenance Test Connector - Pin Assignments

CONNECTOR LOCATION (BLOCK-PIN)	LABEL GROUP PT	DESCRIPTION
A-2	A1*	TEST POINT SELECT
A-9	A2*	
A-15	A3*	
A-14	A4*	
A-3	A5*	
A-4	A6	
A-5	A7	
A-6	A8	
A-10	B1*	
A-13	B2*	
A-17	B3*	
A-20	B4*	
A-7	B5	
A-12	B6	
A-16	B7	
A-22	B8	*Tied to DTM recorder output test jacks.
A-19	SA1	SIGNAL ACCESS TEST JACKS
A-25	SA2	
A-23	SA3	
A-27	SA4	
A-29	SA5	
A-33	SA6	
A-35	SA7	
A-24	SA8	
A-26	SA9	
A-30	SA10	
A-32	SA11	
A-34	SA12	
		DIGITAL I/O
B-1	OBDPF0	Data line 0
B-2	OBDPF1	Data line 1
B-3	OBDPF2	Data line 2
B-4	OBDPF3	Data line 3
B-5	OBDPF4	Data line 4
B-6	OBDPF5	Data line 5
B-7	OBDPF6	Data line 6

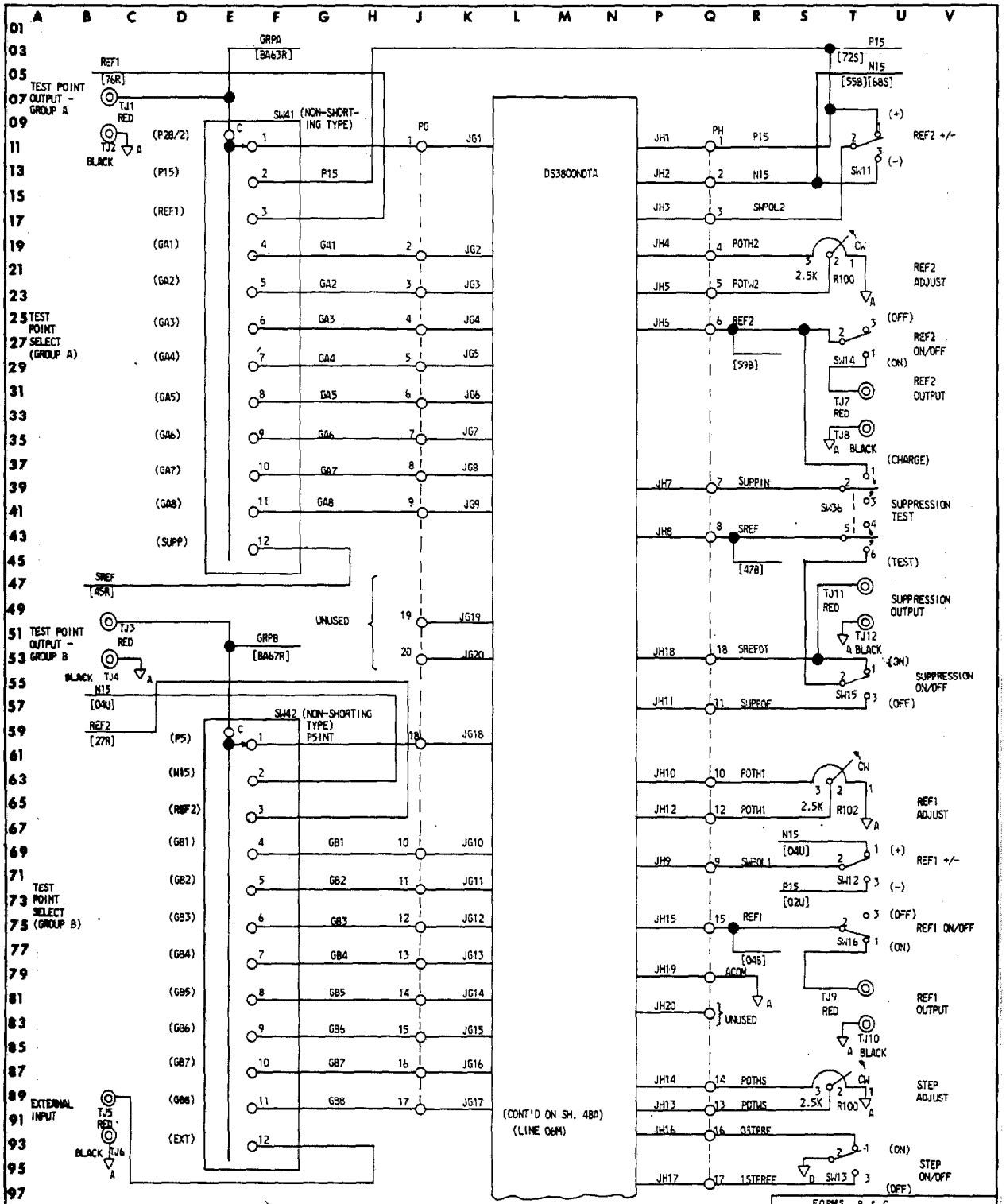
10. APPENDIX (continued)

CONNECTOR LOCATION (BLOCK-PIN)	LABEL GROUP PT	DESCRIPTION
B-8	OBDPF7	Data line 7
B-9	OBDPFP	Data parity
B-10	OBAPF0	Address line 0
B-11	OBAPF1	Address line 1
B-12	OBAPF2	Address line 2
B-13	OBAPF3	Address line 3
B-14	OBAPF4	Address line 4
B-15	OBAPF5	Address line 5
B-16	OBAPF6	Address line 6
B-17	OBAPF7	Address line 7
B-18		Reserved
B-19		Reserved
B-20		Reserved
B-21		Reserved
B-22		Reserved
B-23		Reserved
B-24		Reserved
B-25		Reserved
B-26		
B-27		
B-28		
B-29		
B-30		
B-31		
B-32		
B-33		
		DIGITAL (BUS AND DIRECT I/O)
C-2		Spare
C-11		Spare
C-6		Spare
C-8		Spare
C-10		Spare
C-12		Spare
C-30	OBCFHL	Bus control - Hold
C-32	OBCFHA	Bus control - Hold ack.
		LOGIC OUTPUT LEDS
C-15	1LO8	
C-17	1LO9	
C-19	1LO10	
		LOGIC INPUT SWITCHES
C-14	1L11	
C-16	1L12	
C-18	1L13	
C-20	1L14	
C-22	1L15	

10. APPENDIX (continued)

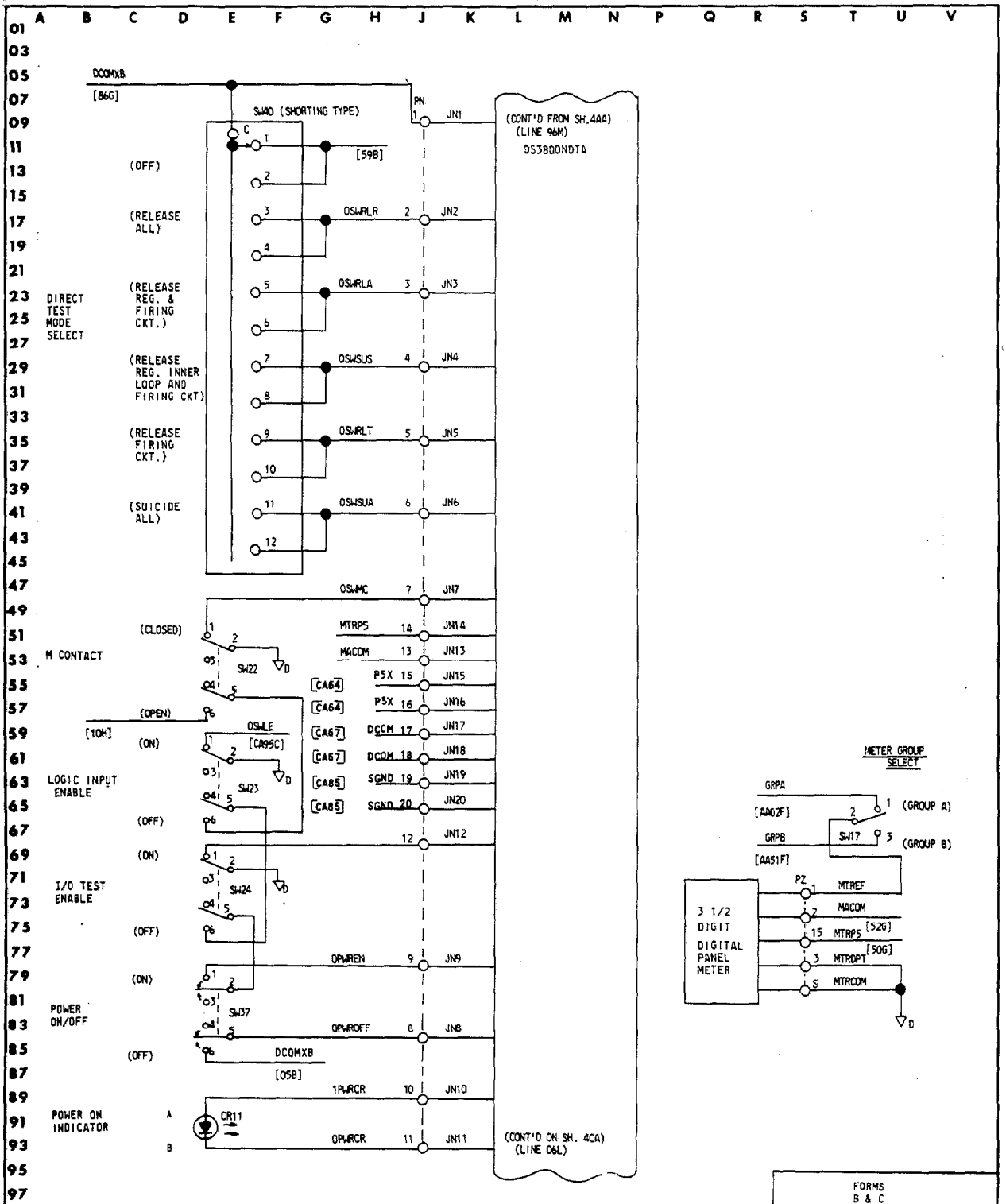
CONNECTOR LOCATION (BLOCK-PIN)	LABEL GROUP PT	DESCRIPTION
C-24	1L16	
C-25	1L17	
C-26	1L18	
C-27	1L19	
C-28	1L10	
C-21	ORSU	Ref. manipulation suicide release
C-23	OMCØP	Motor contactor control (O-Open M)
C-37	OSUA	Suicide reg. and firing circuit
C-38	OSUSX	Release reg. inner loop and firing circuit suicide
C-39	ORLT	Release firing circuit suicide
C-40	ORLA	Release reg. and F.C. suicides
C-35	OMCCL	Motor contactor control (O-Close M)
C-33	P5	Power Bus - P5
C-31	DCOM	Power Bus - DCOM
C-36	DCOM	Power Bus - DCOM
C-29	P5	Power Bus - P5
C-34	P5	Power Bus - P5
C-13	DCOM	Power Bus - DCOM
C-1	ACOM	Power Bus - DVM Ref. Common
C-3	SGND	Power Bus - Analog shield
C-5	ACOM	Power Bus - ACOM
C-7	OMCS	Motor contactor status (O-closed)
C-9	-	Spare
C-4	DCOM	Power Bus - digital shield
		DIGITAL (BUS I/O)
B-34		Spare
B-35		Reserved
B-36		Reserved
B-37	1BCFRT	Bus Control - F Reset
B-38	1BCFRS	Bus Control - Read Strobe
B-39	1BCFDA	Bus Control - Device Ack.
B-40	1BCFOS	Bus Control - Write Strobe
		SUPPRESSION TEST
A-8	SUPPØ	F.C. Suppression test input
A-31	P28	Power bus
A-38	P15	Power bus
A-40	N15	Power bus
A-39	ACØM	Power bus
A-1	1LO1	LOGIC OUTPUT LEDS
A-11	1LO2	
A-18	1LO3	
A-21	1LO4	
A-28	1LO5	
A-36	1LO6	
A-37	1LO7	

DRIVE TEST MODULE



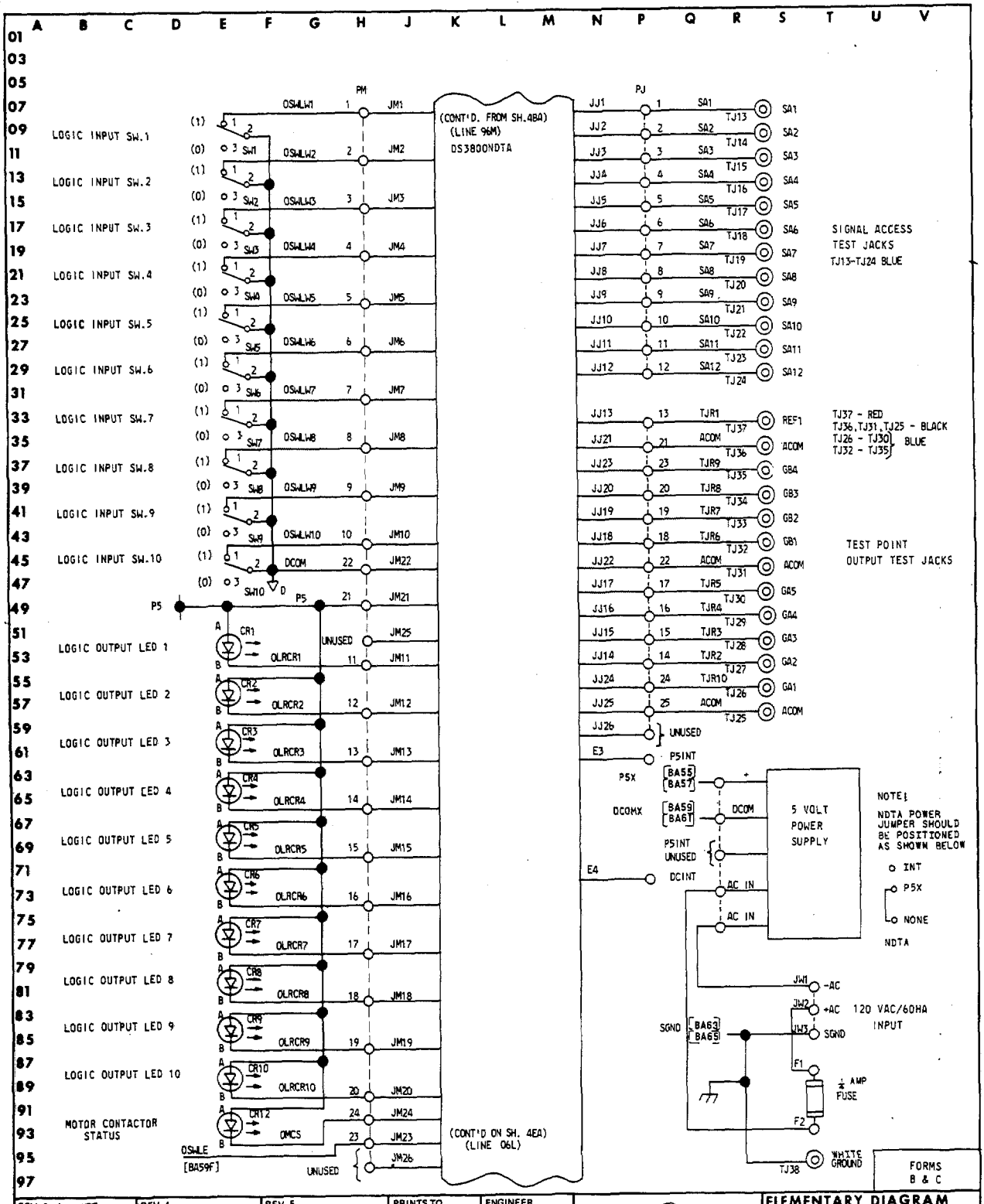
REV.1 AN-OEE Mallard 6-2-78	REV.4	REV.5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	FORMS B & C ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV.2 AN" OEI BLEAVY 2-15-79	ISSUED 4/24/78					DS3820DTM
REV.3 OEL 5/17/79	MADE BY E. Y. LINN					CONT. ON SH. 48A SH. NO. 44A

DRIVE TEST MODULE



REV. 1 AN-OEL Richard 6-2-78	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER [illegible]	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	FORMS B & C ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV. 2 AN OEL B. LEAVY 2-15-79	ISSUED 4/24/79					DS3820DTM
REV. 3 OEL 5/17/79	MADE BY E.Y. LINN					CONT. ON SH. 4CA SH. NO. 4BA

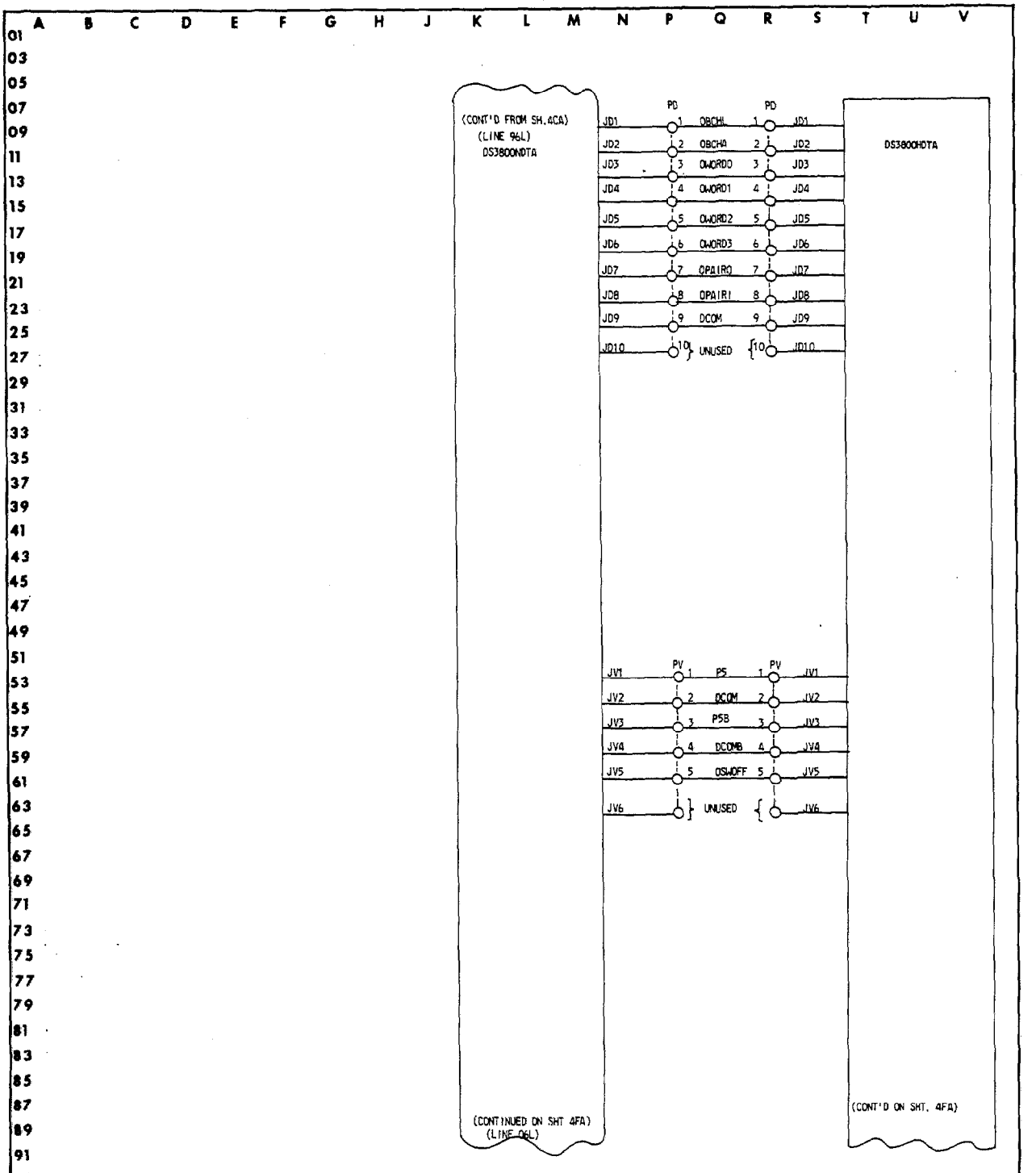
DRIVE TEST MODULE



REV. 1 AN-0E8 W.Reed 6-2-78	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN
REV. 2 AN 0E1 B LEAVY 2-15-79	ISSUED 4/24/78			
REV. 3	MADE BY E.Y. LINN			

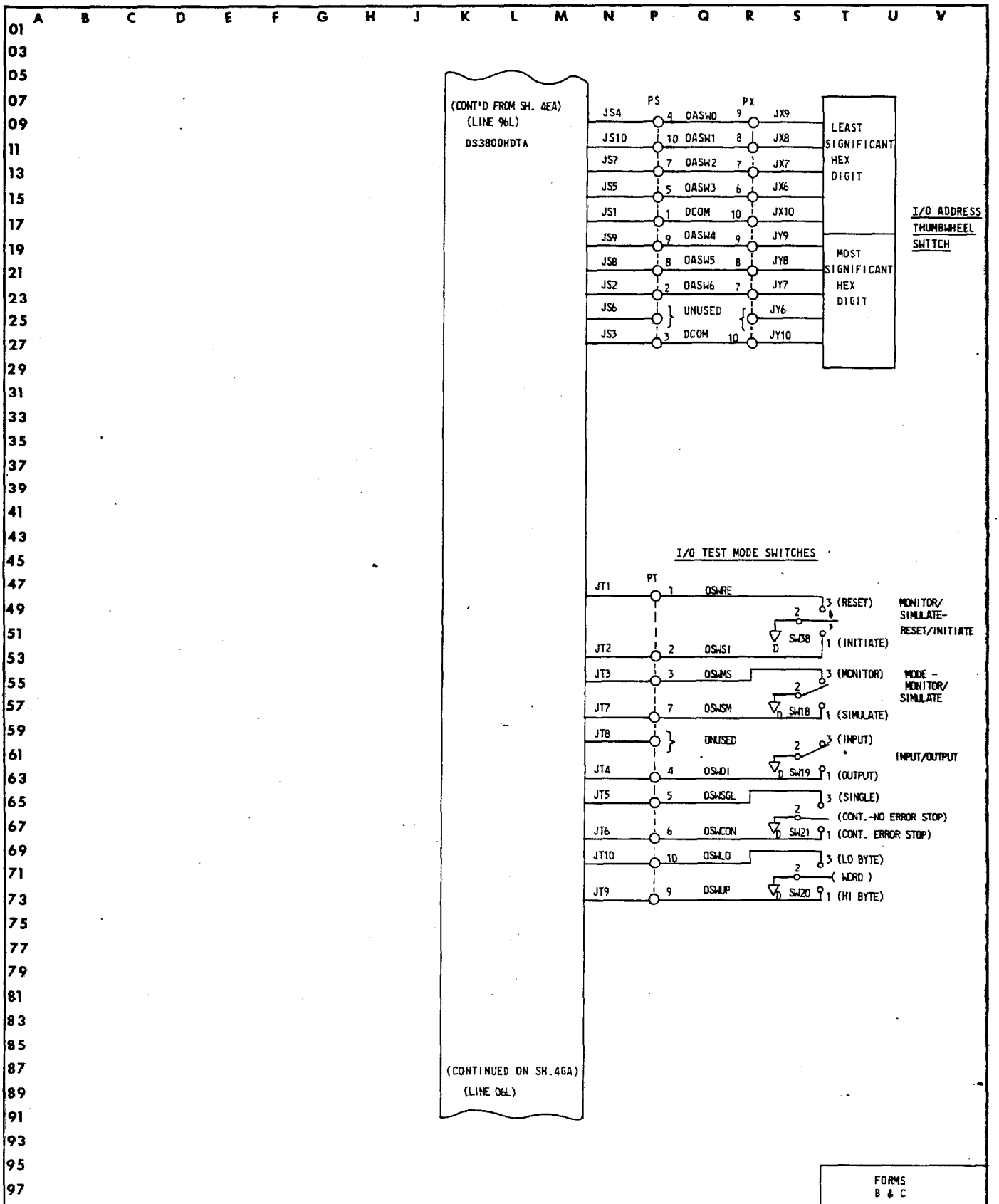
GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
DRIVE TEST MODULE
DS38200TM
CONT. ON SH.4EA SH. NO.4CA



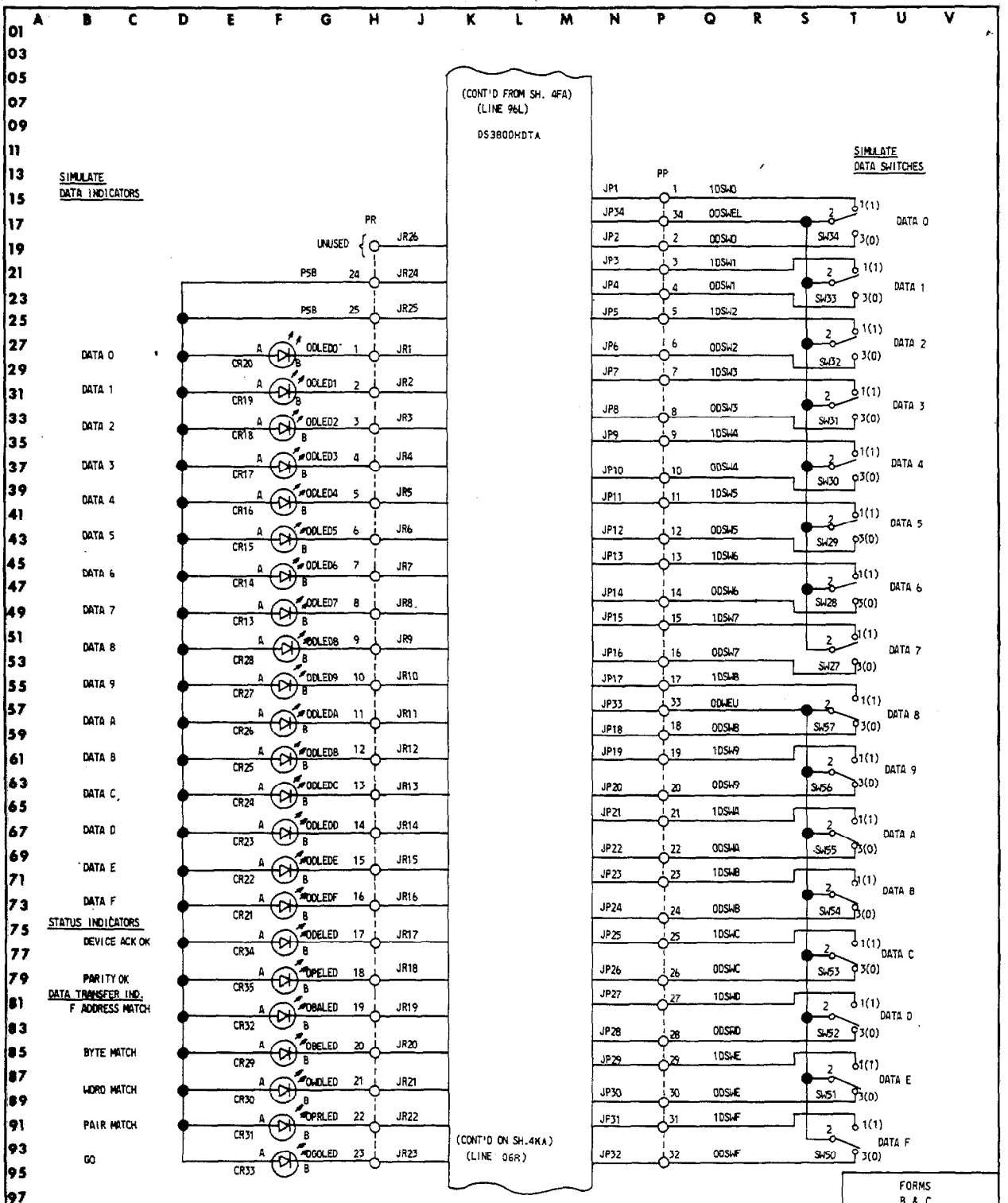
FORMS B & C

REV. 1 AN - 061 & LEAVY 2-15-79	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV. 2	ISSUED 4/24/75					DS3820DTM
REV. 3	MADE BY E. Y. LINN					CONT. ON SH. 4FA SH. NO. 4EA



FORMS B & C

REV.1 AN - OEJ B. LEAVY 2-15-79	REV.4	REV.5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV.2	ISSUED 4/24/78					
REV.3	MADE BY E.Y. LINN					DS3820DTM CONT. ON SH. 4GA SH. NO. 4FA

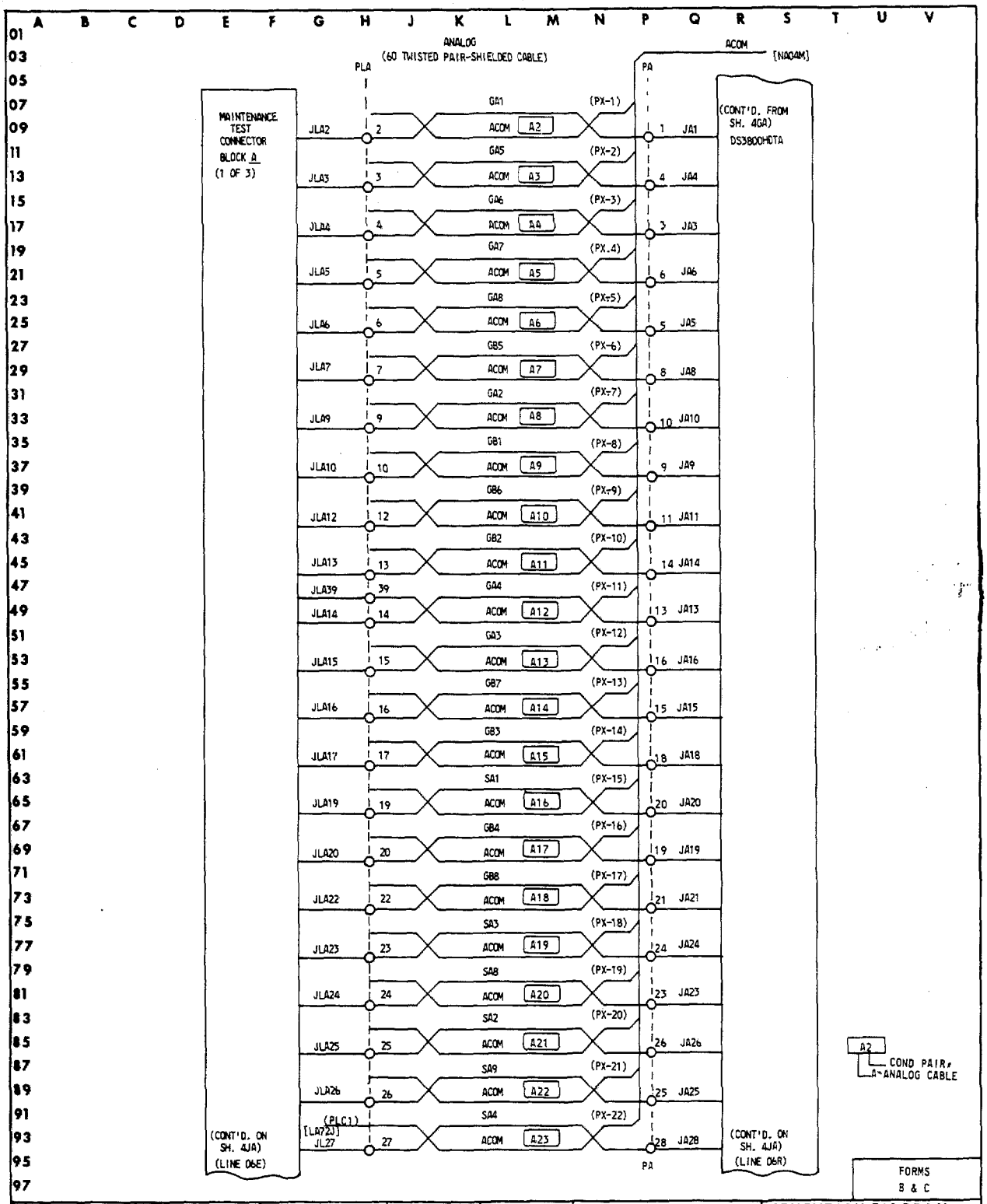


REV. 1 AN-OEE 6-2-78	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN
REV. 2 AN-OEJ E. LEAVY 2-15-79	ISSUED 4/24/78			
REV. 3 OEC 5/17/79	MADE BY E.Y. LINN			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
DRIVE TEST MODULE
DS3820DTM
CONT. ON SH. 4HA SH. NO. 4GA

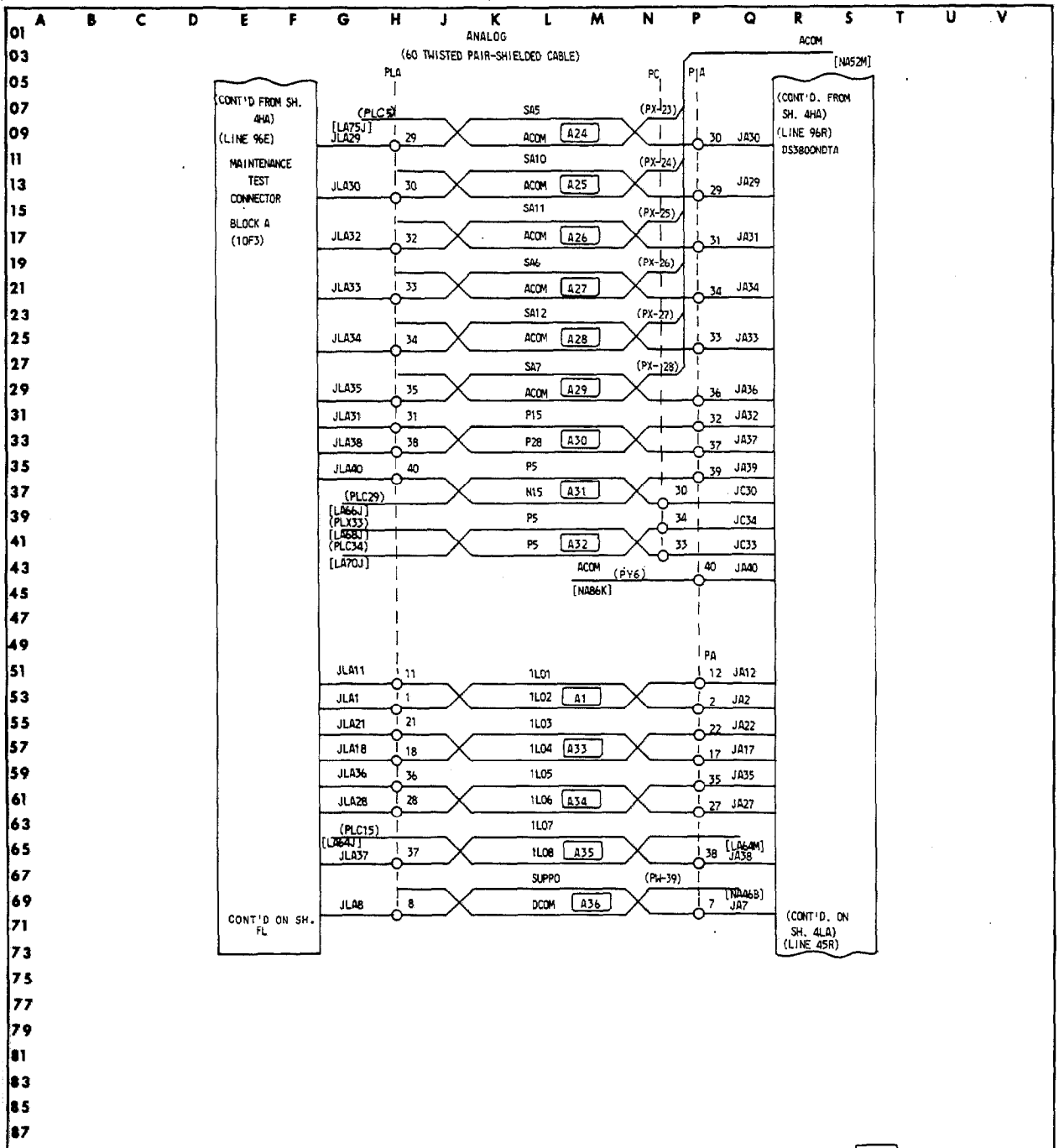
DRIVE TEST MODULE



REV. 1 OEL W.B.K 6/19/79	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2	ISSUED 2/1/79				
REV. 3	MADE BY E.Y. LINN				

ELEMENTARY DIAGRAM
DRIVE TEST MODULE
DS3820DTM
CONT. ON SH. 4JA SH. NO. 4HA

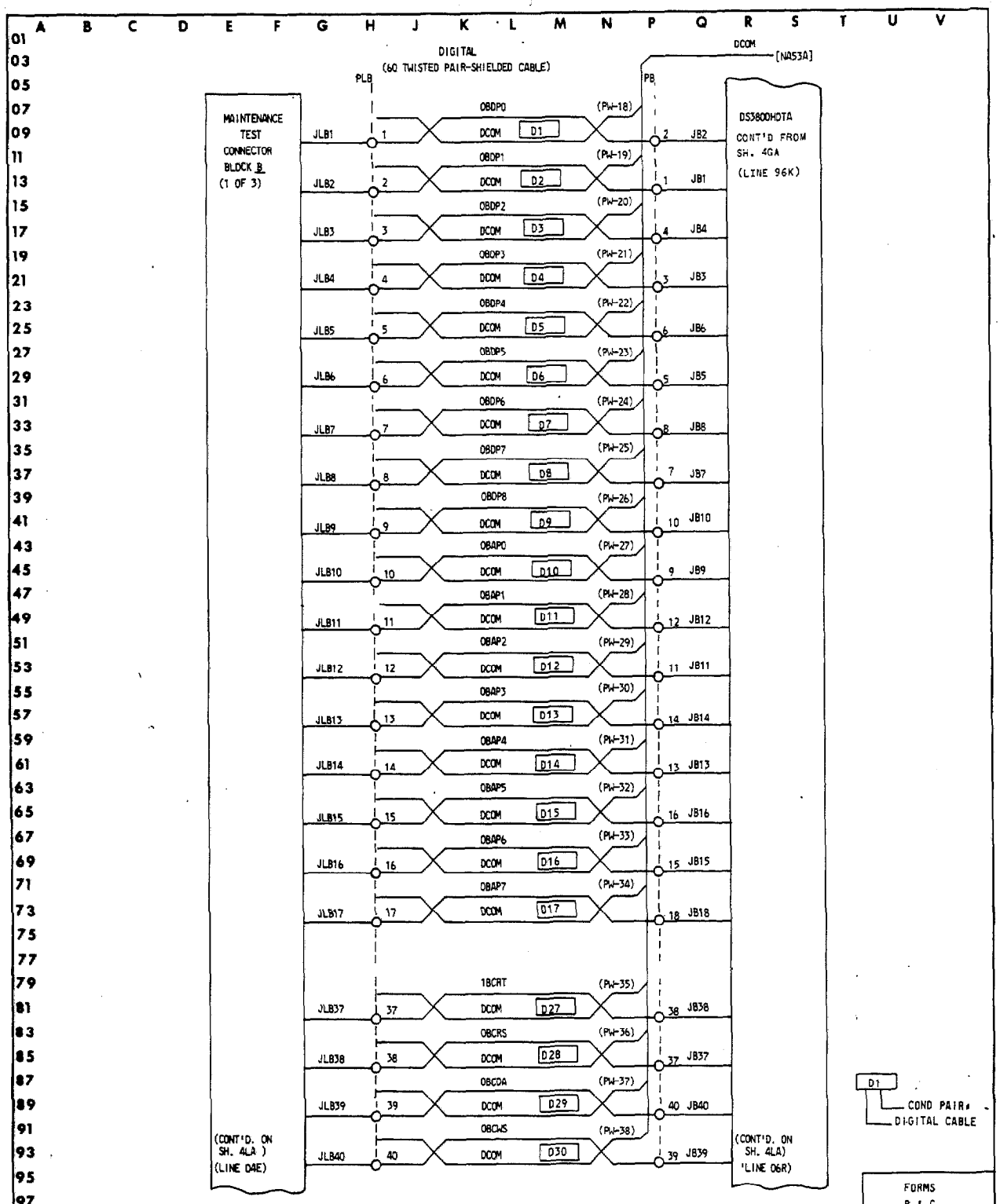
DRIVE TEST MODULE



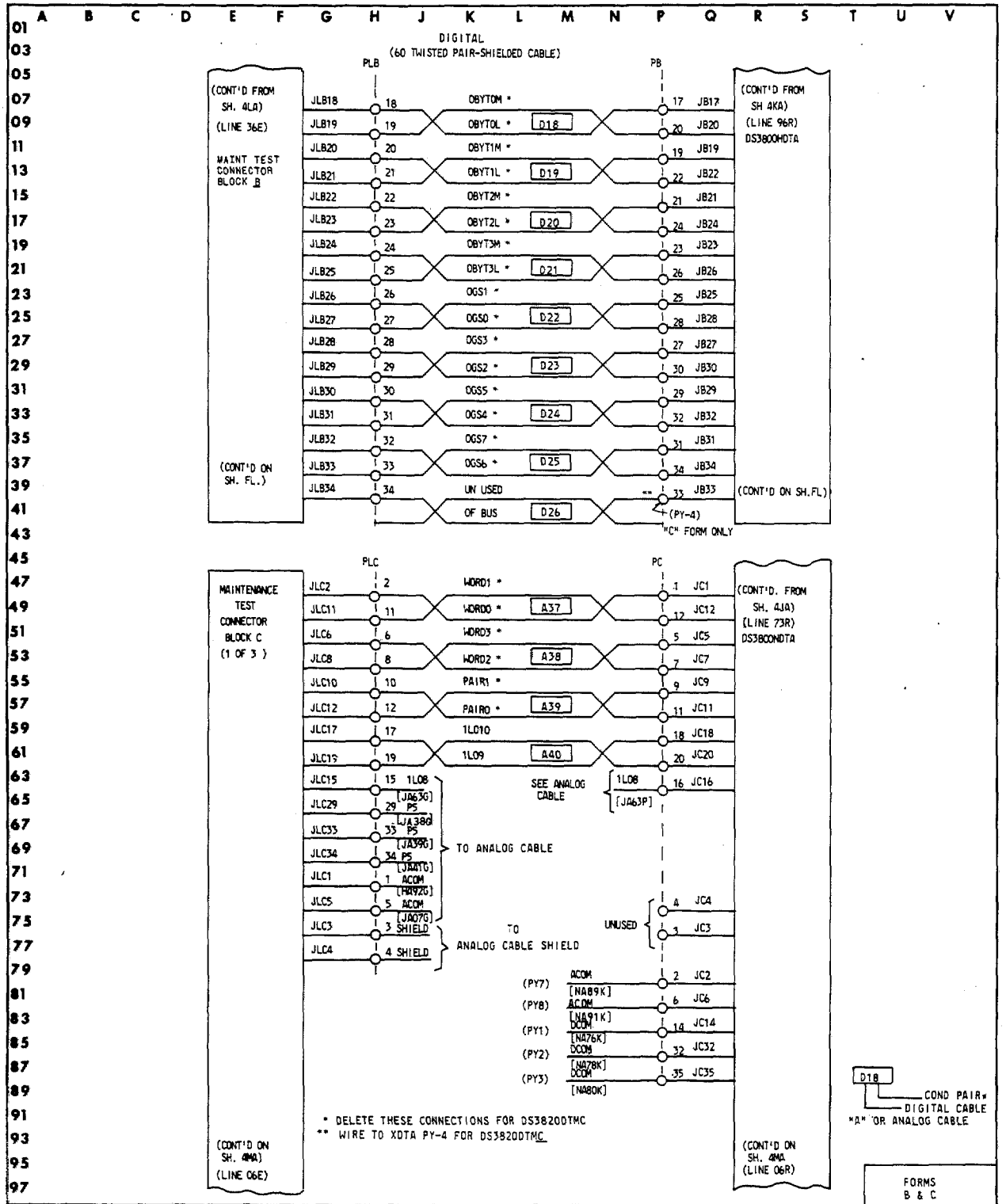
FORM B&C

REV.1 OEL W.B.R. 6/17/79	REV.4	REV.5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV.2	ISSUED 2/11/79					DS38200TM
REV.3	MADE BY E.Y. LINN					CONT. ON SH. 4KA SH. NO. 4JA

DRIVE TEST MODULE

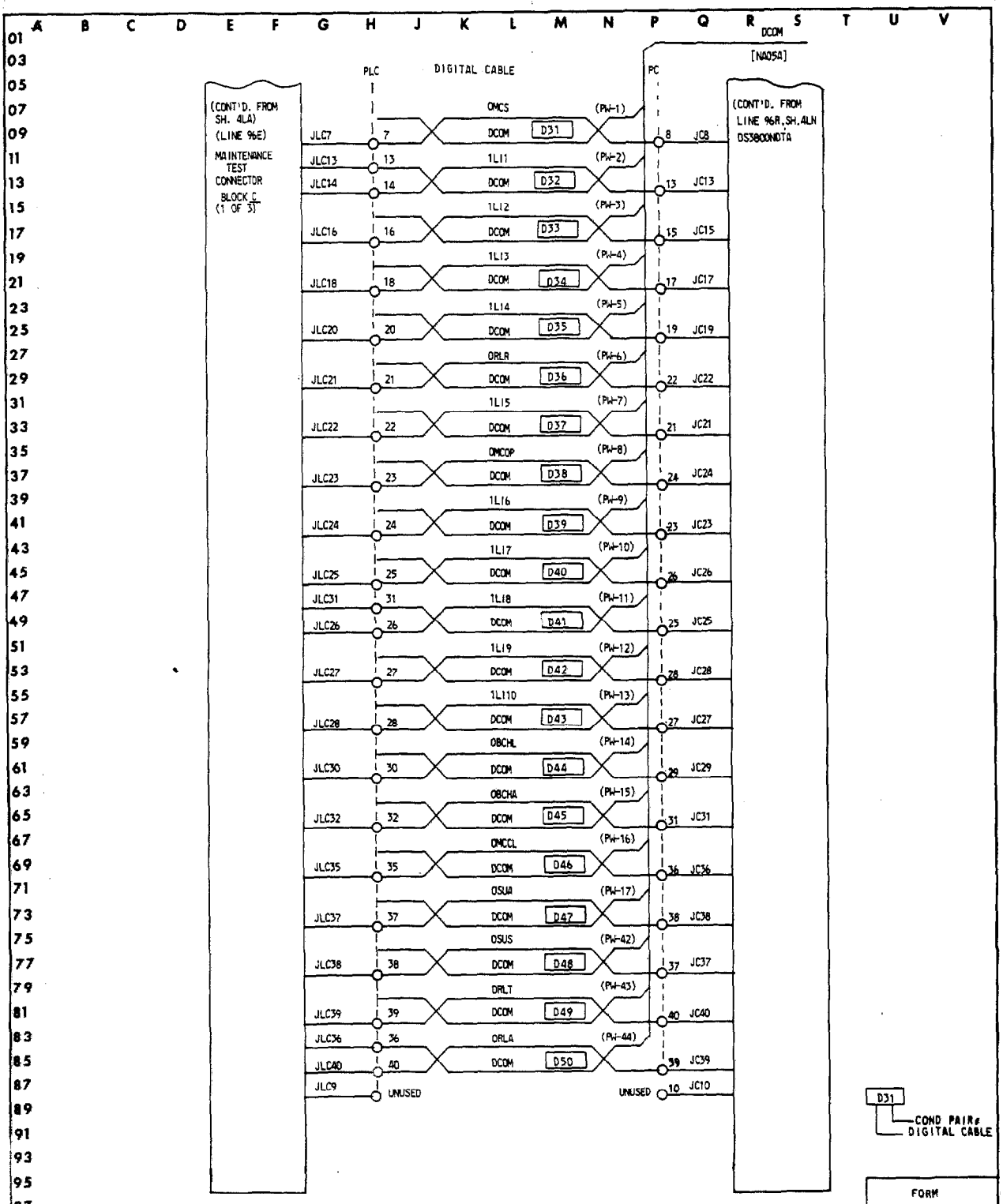


REV. 1: OEL W.B.K 5/17/79	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV. 2	ISSUED 2/11/79					DS3820DTM
REV. 3	MADE BY E.Y. LINN					CONT. ON SH 4LA SH. NO. 4KA



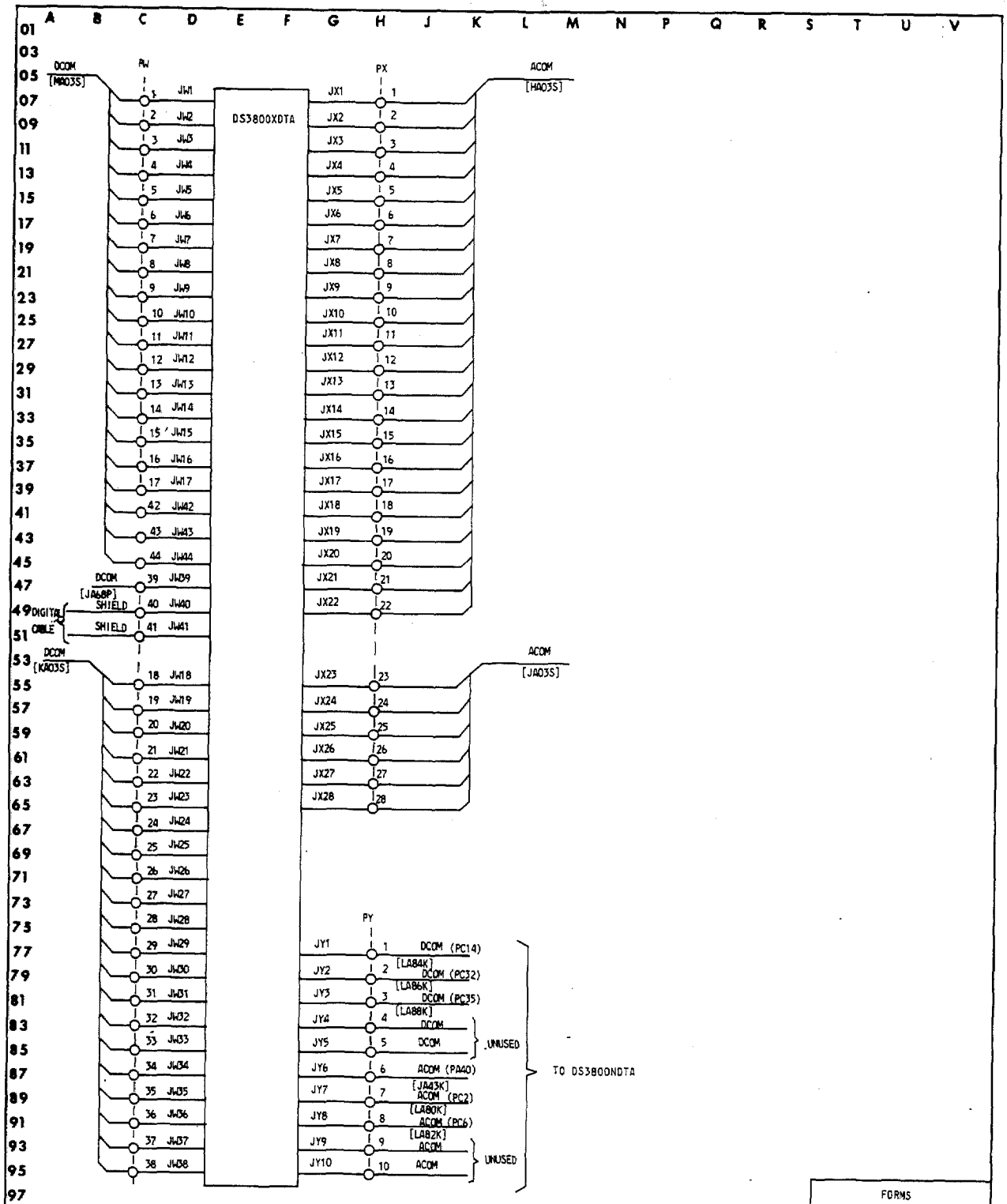
REV. 1 OEL W.B.K. 5/17/79	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV. 2	ISSUED 2/11/79					DS3820DTM
REV. 3	MADE BY E.Y. LINN					CONT. ON SH. 4MA SH. NO. 4LA

DRIVE TEST MODULE



REV. 1 OEL W.B.I 4/17/79	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER Linn	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV. 2	ISSUED 2/11/79					
REV. 3	MADE BY E.Y. LINN E.L.					DS3820DTM CONT. ON SH. 4NA SH. NO. 4MA

DRIVE TEST MODULE



REV. 1 OEL WBK 9.7.79	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER LINN	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DRIVE TEST MODULE
REV. 2	ISSUED 2/11/79					
REV. 3	MADE BY E.Y. LINN					DS3820DTM CONT. ON SH. FL. SH. NO. 4NA

FORMS B & C

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC